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Previsione del carico e della generazione: un'introduzione

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Introduction to Forecasting

Basic Concepts and Tools

*“An approximate answer to the right problem
is worth a good deal more
than an exact answer to an approximate problem”*

Prof. John Tukey

Introduction

Forecasting applied across all segments of the electric power industry: **generation, transmission, distribution, and retail.**

Forecasting applications span power supply planning, transmission and distribution system planning, power system operations and maintenance, financial planning, rate design, and more.

A reliable forecasting capability is essential for **managing the uncertainty** introduced by renewable generation and aligning it with actual demand. This is especially important in the context of active demand response programs, flexibility markets, and load shaping strategies.

Forecasting also contributes to **grid security** by providing critical information to anticipate imbalances and identify vulnerable scenarios in advance.

In a **microgrid** context, accurate forecasting techniques and methodologies are crucial during the **planning, design, and operational phases.**

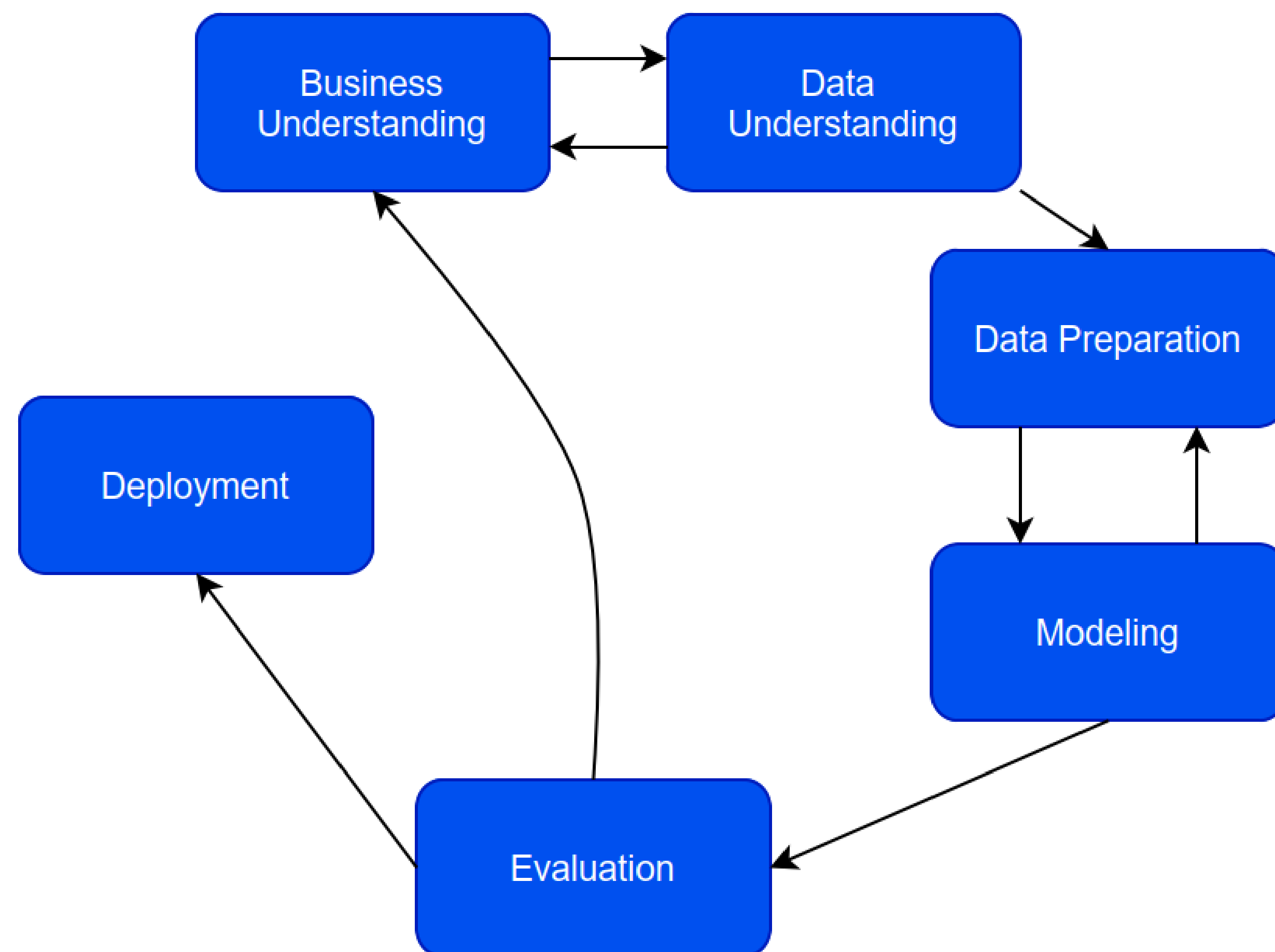
Cross-Industry Standard Procedure for Data Mining

Data-driven forecasting follows standardized procedures for the continuous development, deployment, monitoring, and improvement of models—extending well beyond the energy sector.

One widely adopted framework we will use is the **Cross-Industry Standard Process for Data Mining (CRISP-DM)**.

Ref:

- R. Wirth and J. Hipp, “CRISP-DM: Towards a standard process model for data mining,” 4th International Conference on the Practical Applications of Knowledge Discovery and Data Mining, pp. 29–39, 2000.
- C. Schröer, F. Kruse, and J. Marx Gomez, “A systematic literature review on applying crisp-dm process model,” *Procedia Computer Science*, vol. 181, pp. 526–534, 2021.



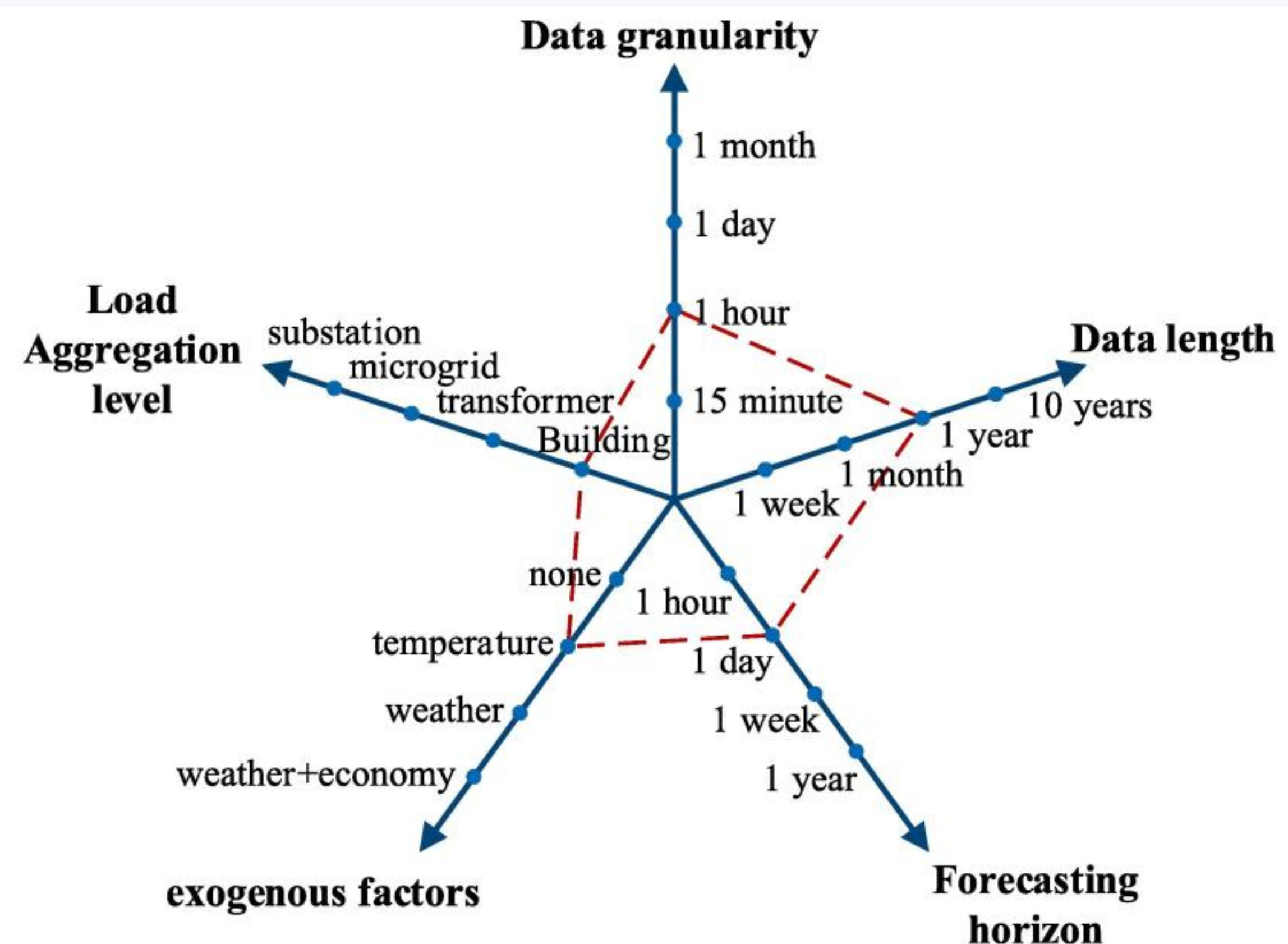
Business understanding

The forecasting problem, in particular, can be framed in various ways, as the quantities to be forecasted depend on the business problem specifics.

For example, the load forecasting problem can take very different characteristics: **data granularity** (i.e., the frequency of the data) and **forecasting horizon** (i.e., how far in time must the model forecast), in particular, are very important in defining the subsequent CRISP-DM steps.

Ref:

- Y. Li, S. Zhang, R. Hu, and N. Lu, "A meta-learning based distribution system load forecasting model selection framework," *Applied Energy*, vol. 294, p. 116991, 2021.



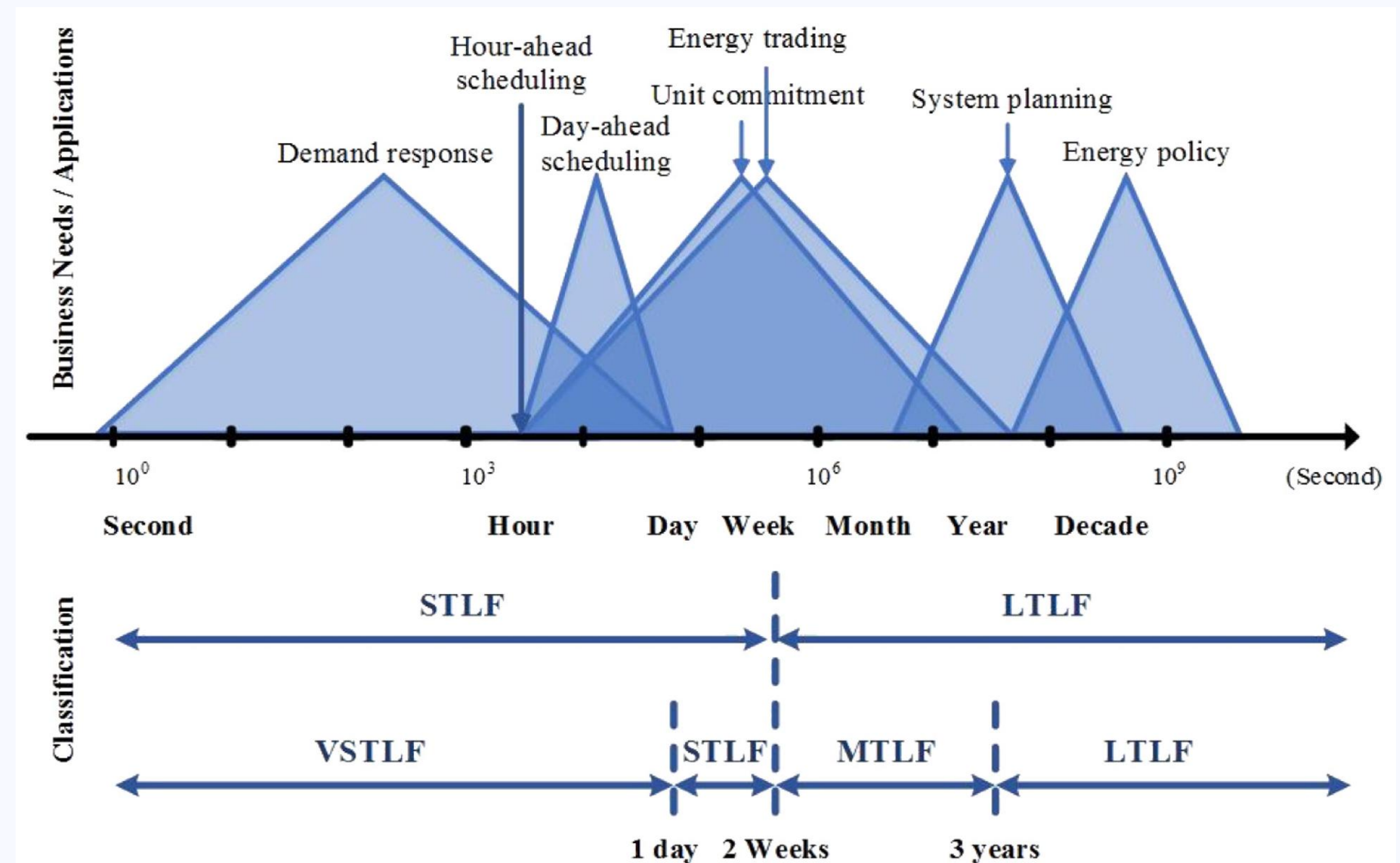
Business understanding – Forecasting classification

It is possible to classify the load forecasting problems according to their **forecast horizon**.

A more liberal classification is to use the two weeks horizon as the cut-off point to separate STLF and LTLF. The main justification for this cut-off is that temperature, an important driving factor of load, can be hardly predicted beyond two weeks.

Ref:

- T. Hong, S. Fan, “Probabilistic electric load forecasting: A tutorial review”, *International Journal of Forecasting*, vol. 32, no. 3, 2016, pp. 914-938
- T. Hong, P. Pinson, Y. Wang, R. Weron, D. Yang, H. Zareipour, “Energy forecasting: A review outlook”, *IEEE Open Access Journal of Power and Energy*, vol. 7, 2020, pp. 376-388



Business understanding

Another important aspect is the **level of aggregation** for the forecasting task.

Indeed, a quantity (such as load) can be split into sub-quantities (typically several sub-loads plus losses), and the question is whether it is necessary to forecast the sub-quantities or it is sufficient to forecast the principal quantity. Indeed, typically **the higher the aggregation level, the easier the forecasting tasks**, both in terms of forecasting KPIs and in terms of complexity of the solution. On the other hand, **a disaggregated forecast delivers more information**, and if the sub-quantities at different levels can be reconciled, the accuracy of the forecast can increase.

Also, the notion of "good" prediction must be defined, both in terms of meaningful **Key Performance Indicators (KPIs)** and in terms of the magnitude of a satisfying predictive analytics solution. In power systems, a small percentage of error can lead to several hundred dollars loss for a utility financial bottom line.

Ref:

- T. Hong, "Crystal ball lessons in predictive analytics," *EnergyBiz*, vol. 12, no. 2, pp. 35–37, 2015.

Business understanding

Finally, since perfect predictions are impossible to achieve, it is necessary to determine whether a **measure of uncertainty associated with the forecasts** is required.

If such a measure is provided, the forecasting is referred to as **probabilistic**. Otherwise, the task is considered a **point forecasting problem**.

In many aspects of the forecasting task, the specific characteristics of the business problem play a crucial role. For example, the forecast origin time, along with constraints on inference and training resources—both directly tied to the available budget in terms of time and money—are important factors to consider.

Ref:

- N. Amjady and M. Hemmati, “Energy price forecasting - problems and proposals for such predictions,” *IEEE Power and Energy Magazine*, vol. 4, no. 2, pp. 20–29, 2006.

Data understanding

Once the type of predictive analytics task is identified and the corresponding data is collected, a closer look at the available information must be given.

Moreover, additional care must be taken to understand whether the data has some **biases** that must be taken care.

Another important aspect to consider is the **quantity and type of data**, as many modeling techniques perform differently depending on the amount of data available.

Although there are exceptions, in general, more data leads to better performance.

Moreover, specific preprocessing techniques are required for different data types.

Ref:

- *W. A. Yousef and S. Kundu, “Learning algorithms may perform worse with increasing training set size: Algorithm-data incompatibility,” Comput. Stat. Data Anal., vol. 74, p. 181–197*

Data understanding

Identifying **bad data** is a crucial part of this phase.

The concept of "bad data" is relative to the specific question being addressed: a data instance considered poor for one task may be perfectly acceptable for another.

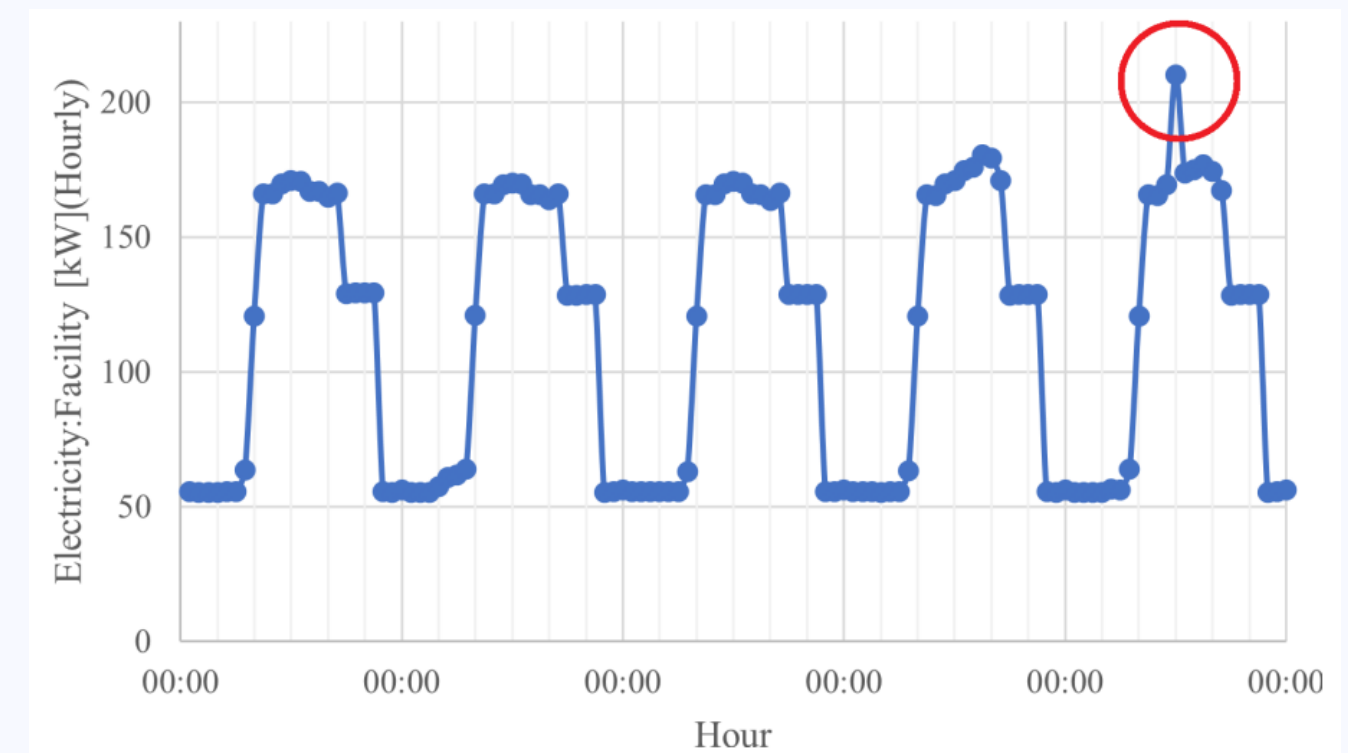
Measurement errors in input data can significantly affect the estimation of causal relationships, while having less impact on the prediction of the response variable itself.

Another way in which data can be bad is if data is not up-to-date, meaning that the model developed may not be relevant for deployment.

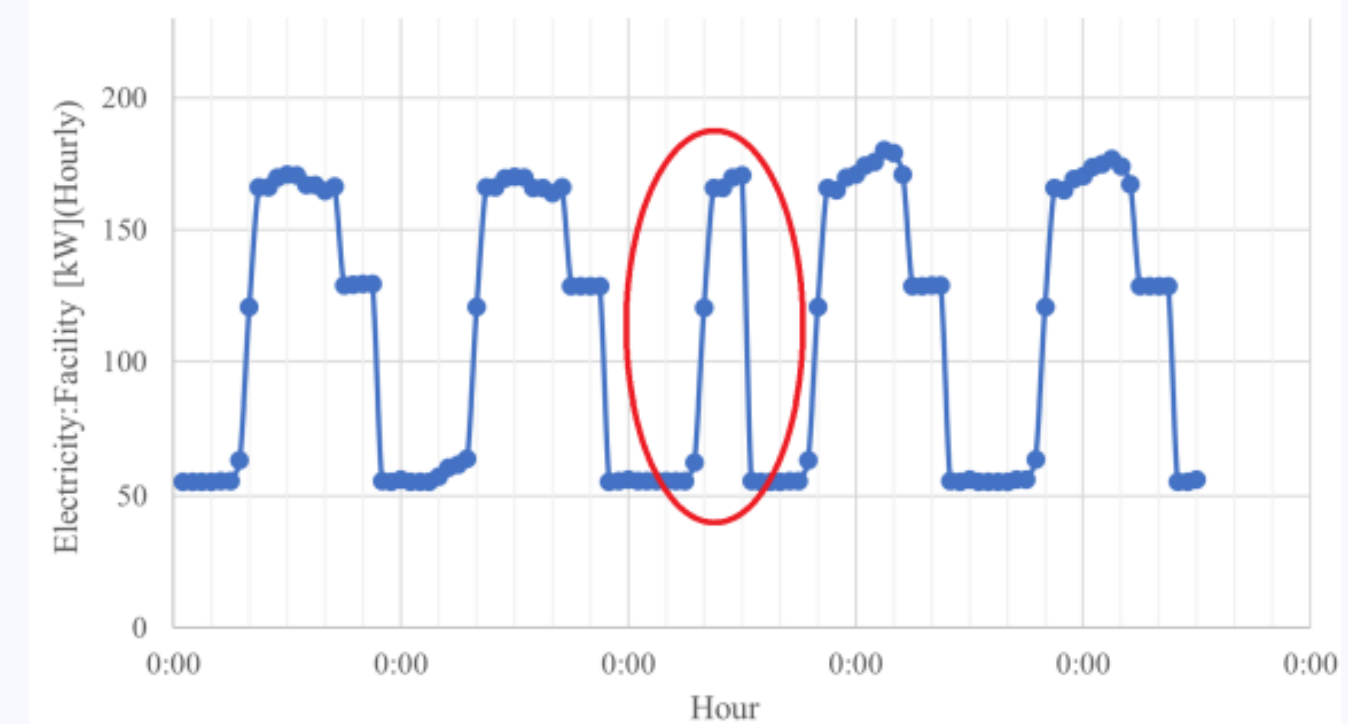
It is not trivial to identify bad data, as expert knowledge and intuition are often necessary.

Ref:

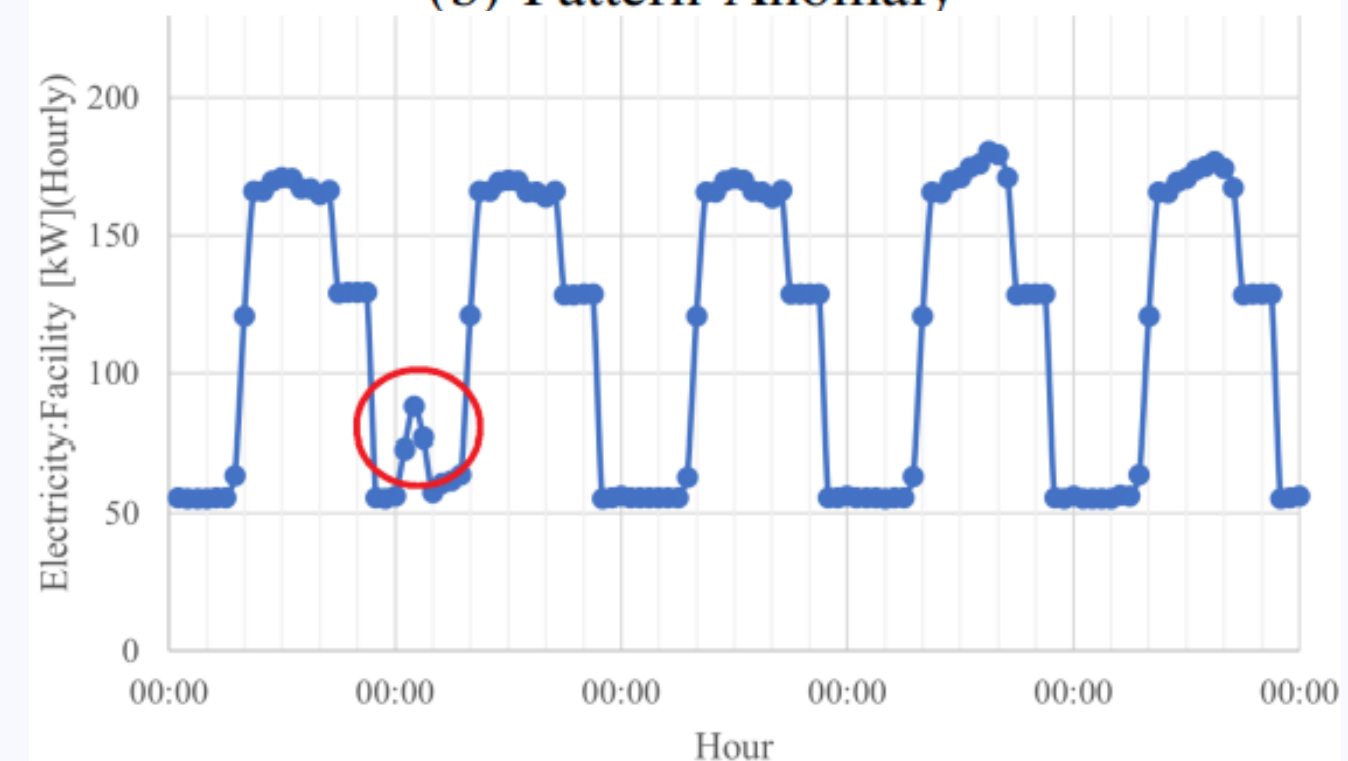
- R. D. D. Veaux and D. J. Hand, "How to Lie with Bad Data," *Statistical Science*, vol. 20, no. 3, pp. 231 – 238, 2005



(a) Point Anomaly



(b) Pattern Anomaly



(c) Contextual Anomaly

Data preparation – Missing data

Missing data can mean that an entire record is absent or that individual features of a record are missing. It is very common in real-world datasets and can severely hinder the performance of the desired models.

There are several types of missing data:

- **Missing Completely At Random (MCAR)** when the probability of a data point being missing is completely independent of any variables in the dataset;
- **Missing At Random (MAR)** when it is independent of its own (missing) value but probabilistically dependent on the value of another variable;
- **Missing Not At Random (MNAR)** when it is dependent on its own (missing) value.

Ref:

- R. Tawn, J. Browell, and I. Dinwoodie, “Missing data in wind farm time series: Properties and effect on forecasts,” *Electric Power Systems Research*, vol. 189, p. 106640, 2020.

Data preparation – Missing data

There are several ways to handle missing data:

- **Complete Case Analysis**, where records with missing values are discarded. This approach can significantly reduce the dataset size and may lead to biased results if the data is Missing at Random (MAR) or Missing Not at Random (MNAR).
- **Imputation**, where missing values are replaced with simple substitutes (e.g., the last available value, interpolation, regression, etc.). However, this method becomes less effective when the proportion of missing data is high.

Other issues arise that deserve careful attention in developing a forecasting solution:

- **Non-stationary data** (mean and variance change over time)
- **Seasonality**
- **Peaks and high volatility**

Ref:

- R. Tawn, J. Browell, and I. Dinwoodie, “Missing data in wind farm time series: Properties and effect on forecasts,” *Electric Power Systems Research*, vol. 189, p. 106640, 2020.

Data preparation – Feature extraction

After bad, missing, and anomalous data have been filtered out, the next step is to construct inputs for predictive models—a process known as **feature extraction**.

Even in the absence of exogenous variables, the **selection of features** (i.e., functions derived from the original data) has a significant impact on **the performance of forecasting techniques**.

Initially, features are computed from the data, followed by a selection process. These two steps are referred to as **feature extraction** and **feature selection**, respectively.

Various techniques for feature extraction include:

- Feature learning (e.g. with neural networks)
- Signal processing
- Local features methods (i.e., moving average)
- Linear and non-linear feature transformations
- Non-linear expansions
- Feature discretization
- Normalization and standardization technique

Ref:

- *I. Guyon and A. Elisseeff, An Introduction to Feature Extraction, pp. 1–25. Berlin, Heidelberg: Springer Berlin Heidelberg, 2006.*

Data preparation – Feature selection

Feature selection presents the challenge of **retaining “relevant” features** while **discarding irrelevant ones**—a process that is highly dependent on the specific problem, domain, and dataset.

The notion of relevance itself is closely tied to the forecasting objective: what may be irrelevant for short-term forecasting could be crucial for long-term forecasting, and vice versa.

Feature selection methods produce a subset of the original features or rank all the features. There are primarily four types of **feature selection techniques**:

- Filter methods
- Wrapper methods
- Embedded methods
- Ensemble of the above methods

All these steps have been referred to as “methodologies” in order to distinguish them from **the actual predictive models, which are referred to as “techniques”**. The distinction is helpful because one can use the same exact predictive technique with many different methodologies and arrive at very different accuracy levels and conclusions.

Ref:

- *R. V. Bolon-Canedo and A. Alonso-Betanzos, “Ensembles for feature selection: A review and future trends,” Information Fusion, vol. 52, pp. 1–12, 2019.*

Modeling

The possibilities for selecting an appropriate forecasting algorithm (“model”) are virtually endless. **The “best” technique depends on the specific task, domain, and dataset, making it unrealistic to expect a single “optimal” method that works universally across all tasks**—or even across all instances of a single task, such as load forecasting.

A broad range of choices is available when selecting forecasting approaches, including the following:

- **Structured vs. Unstructured Approaches**
- **Physical vs. Statistical Approaches**
- **Global vs. Local Approaches**
- **Integral vs. Decomposition Approaches**

Ref:

T. Januschowski, J. Gasthaus, Y. Wang, D. Salinas, V. Flunkert, M. Bohlke-Schneider, and L. Callot, “Criteria for classifying forecasting methods,” International Journal of Forecasting, vol. 36, no. 1, pp. 167–177, 2020

Modeling – Parameter/Hyperparameter selection

Moreover, each model has a number of **parameters** governed by one or more **hyperparameters**.

There are many hyperparameters selection techniques:

- **Grid Search**, also known as full factorial design.
- **Random Search** is a simple alternative, which samples configurations at random until a specific budget (such as number of evaluations or time) is exhausted.
- **Genetic Search methods**. They can be thought of as guided random search methods.
- **Bayesian Optimization**. These approaches are black-box global optimization iterative algorithms with two key ingredients: a probabilistic surrogate model and an acquisition function to decide which point to evaluate next

Ref:

M. Hibon and T. Evgeniou, “To combine or not to combine: selecting among forecasts and their combinations,” International Journal of Forecasting, vol. 21, no. 1, pp. 15–24, 2005

Modeling – Ensemble?

Finally, a decision must be made on whether to use **one predictive model** or an **ensemble of predictive models**.

Strategies for ensembling: **bagging, boosting, stacking**.

Ensembling is particularly useful when individual forecasts are either comparable in performance or exhibit diversity.

However, ensembles tend to perform poorly when the data is highly volatile or contains a significant number of outliers, as they may learn suboptimal patterns from these anomalies.

In general, the strength of ensembling does not lie in consistently outperforming the best individual model, but rather in reducing the risk of producing poor forecasts.

Ref:

F. Atiya, “Why does forecast combination work so well?,” International Journal of Forecasting, vol. 36, no. 1, pp. 197–200, 2020.

CRISP-DM phase	Subphase	Options
Data Preparation	Bad Data Management	Complete Case Analysis; Imputation; outlier removal
	Feature Extraction	Feature Learning; Signal enhancement; local feature methods; feature transformation; non-linear expansions; discretizations; normalizations; standardizations; etc.
	Feature Selection	Filter; Wrapper; Embedded methods.
	Selection Ensembles	Homogeneous; Heterogeneous.
Modeling	Choice of Technique	Structured or Unstructured; Physical or Statistical; or Hybrid.
	Approach	Global or Local; Integral or Decomposition; MIMO; recursive; or direct.
	Hyperparameter Selection	Grid Search; Random Search; Genetic Search; or Bayesian Optimization.
	Ensemble Method	Bagging; Boosting; Stacking; others.

Evaluation

This can only be done using either a **hold-out validation set** or a **cross-validation procedure**. While cross-validation is the preferred method due to its robustness, it requires running the same model multiple times, which can be computationally intensive.

There are several versions of **cross-validation**:

- **Fixed-Origin**
- **Rolling-Origin-Recalibration**
- **Rolling-Origin-Update**
- **Rolling-Window-Evaluation**

An alternative method is given by the techniques for which the **Akaike Information Criterion (AIC)** or the **Bayesian Information Criterion (BIC)** can be computed.

Ref:

C. Bergmeir and J. M. Benitez, “On the use of cross-validation for time series predictor evaluation,” *Information Sciences*, vol. 191, pp. 192– 213, 2012.

Evaluation – KPI/Benchmark

Selecting a **meaningful error metric** (e.g., MAPE, MAE) **is essential**, and **benchmarking the model** against simple, trivial, state-of-the-art, or business-as-usual models is fundamental for evaluating its effectiveness.

This helps to understand the potentiality of the selected models. It is also important to be fair in comparison by devoting the same resources (i.e., computing time) to each competing model.

In the event of an **unsuccessful evaluation, restarting the cycle becomes necessary**. This allows for a better understanding of the business and data, facilitating the preparation and construction of improved models until a satisfactory evaluation is achieved

Ref:

P. C. Emiliano, M. J. Vivanco, and F. S. de Menezes, “Information criteria: How do they behave in different models?,” Computational Statistics Data Analysis, vol. 69, pp. 141–153, 2014

Deployment

Once the model has received a positive evaluation, the **deployment** phase begins. During deployment, ongoing performance monitoring becomes crucial. This step is essential to determine whether the model continues to function correctly or if there are signs of concept drift—or persistent missing data—that may require retraining the model.

This consideration is especially important for online learning methods, which are capable of adapting to new patterns but may do so at the cost of forgetting previously learned relationships.

Ref:

M. Barque, S. Martin, J. E. N. Vianin, D. Genoud, and D. Wannier, “Improving wind power prediction with retraining machine learning algorithms,” International Workshop on Big Data and Information Security (IWBIS), pp. 43–48, 2018

Short-Term Load Forecasting

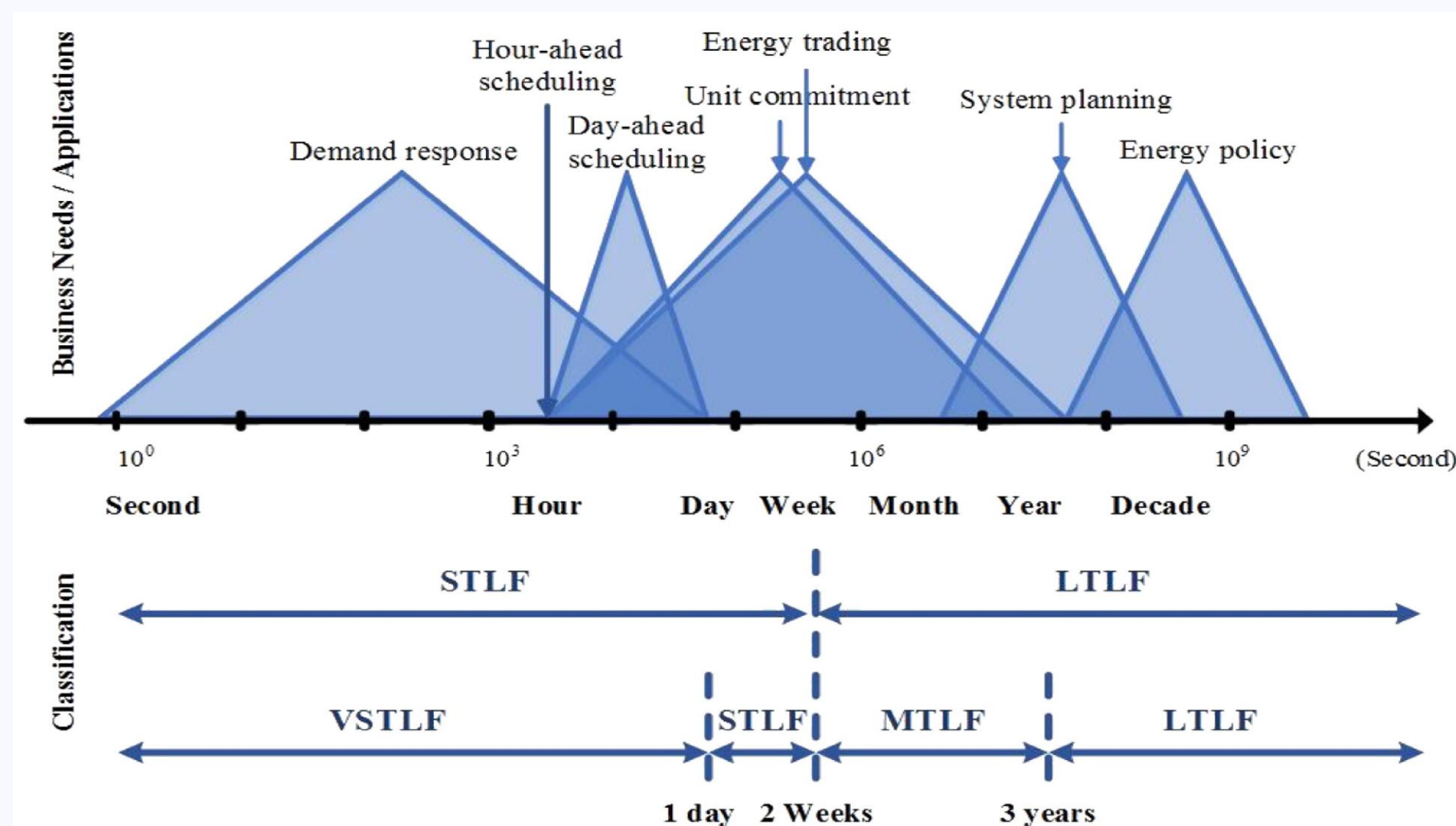
Classification, Load Types, Key Inputs

Classification of load forecasting problem

It is possible to classify the load forecasting problems according to their forecast horizon into the following categories:

- **Very Short Term Load Forecasting (VSTLF);**
- **Short Term Load Forecasting (STLF);**
- Medium Term Load Forecasting (MTLF);
- Long Term Load Forecasting (LTLF).

A more liberal classification is to use the two weeks horizon as the cut-off point to separate STLF and LTLF. The main justification for this cut-off is that temperature, an important driving factor of load, can be hardly predicted beyond two weeks.



Ref:

T. Hong, S. Fan, "Probabilistic electric load forecasting: A tutorial review", *International Journal of Forecasting*, vol. 32, no. 3, 2016, pp. 914-938

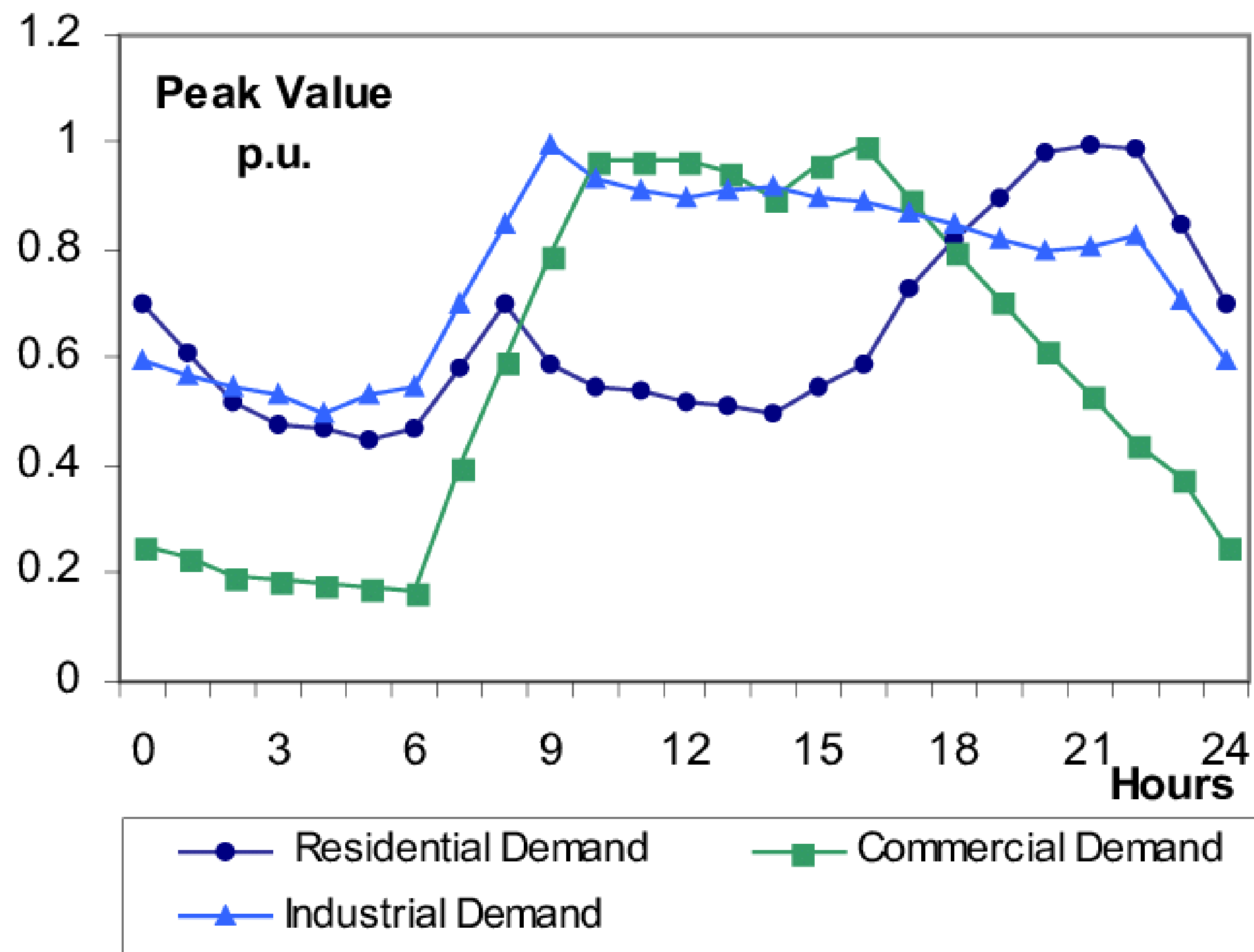
T. Hong, P. Pinson, Y. Wang, R. Weron, D. Yang, H. Zareipour, "Energy forecasting: A review outlook", *IEEE Open Access Journal of Power and Energy*, vol. 7, 2020, pp. 376-388

Classification of load forecasting problem

Horizon	Studies Number	Papers Number	Distribution Percentage
Very Short Term	5	1	2.6%
Short Term	40	17	43.6%
Mid Term	9	3	7.7%
Long Term	58	24	61.5%

Ref: C. Kuster, Y. Rezgui, M. Mourshed, "Electrical load forecasting models: A critical systematic review", Sustainable Cities and Society, vol. 35, 2017, pp. 257-270

Load types



Key inputs – Short term load forecasting

- Time of day
- day of the year
- day of the week
- type of day (holiday/weekday)
- historical load data
- weather variables (temperature, humidity, etc.)

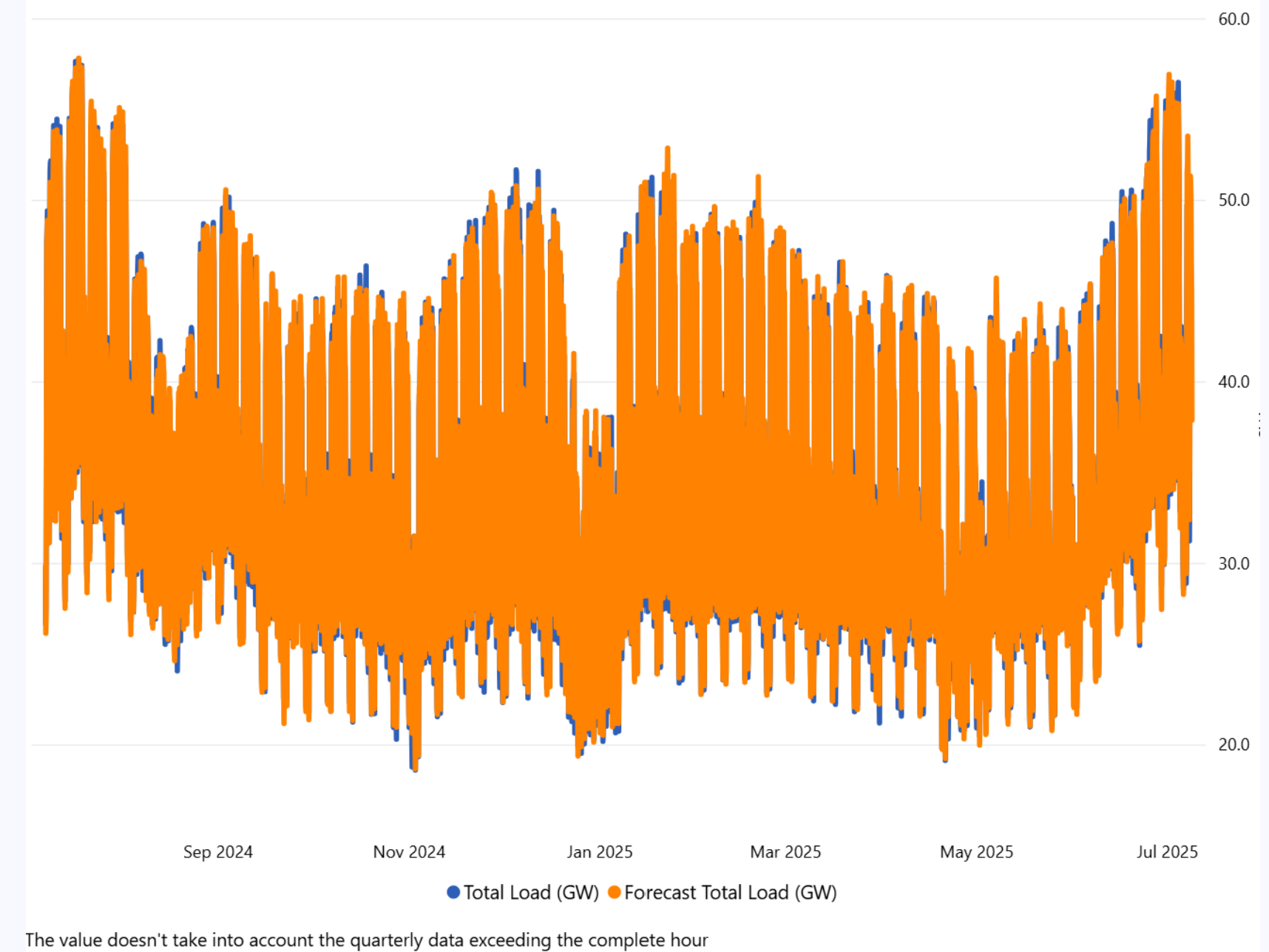
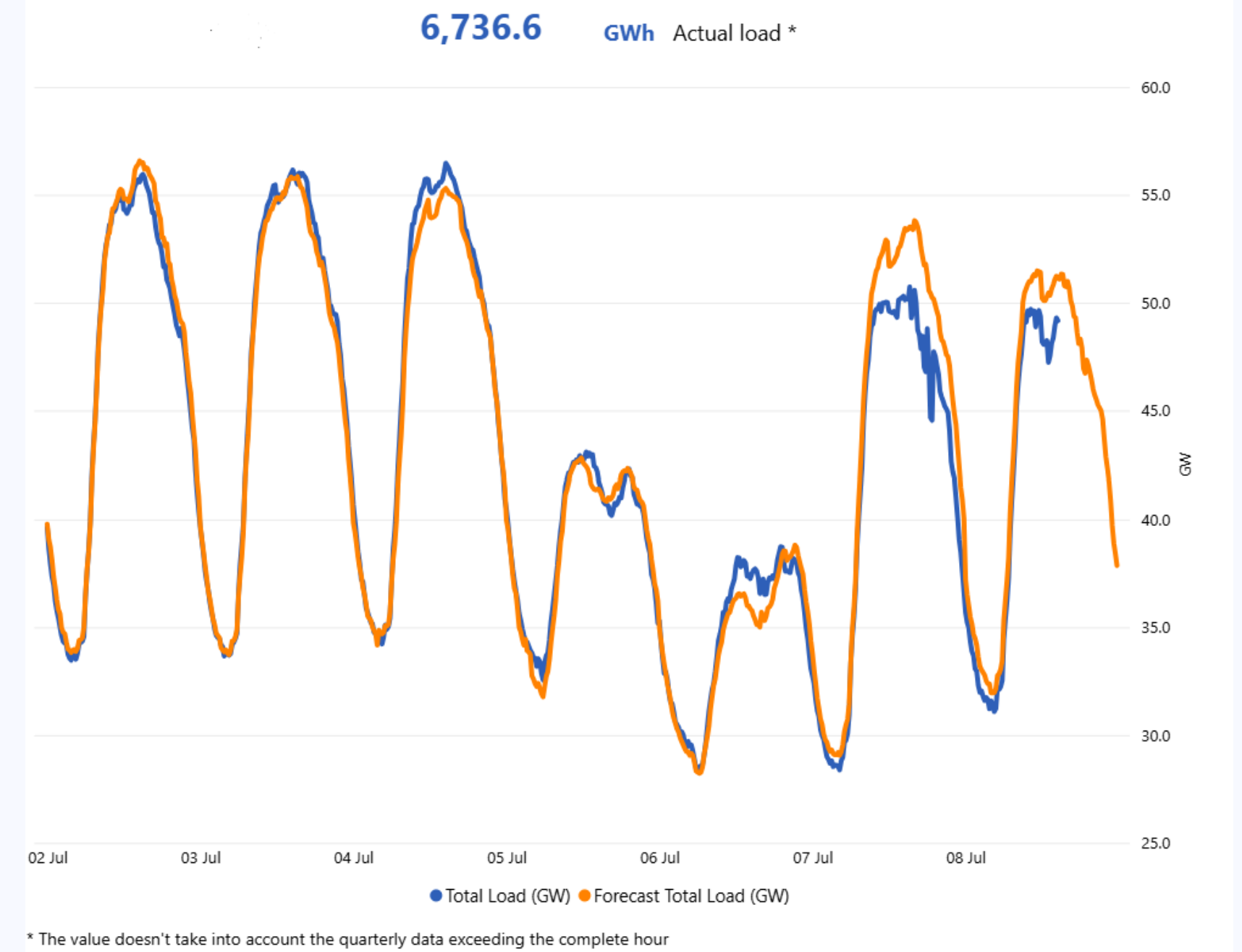
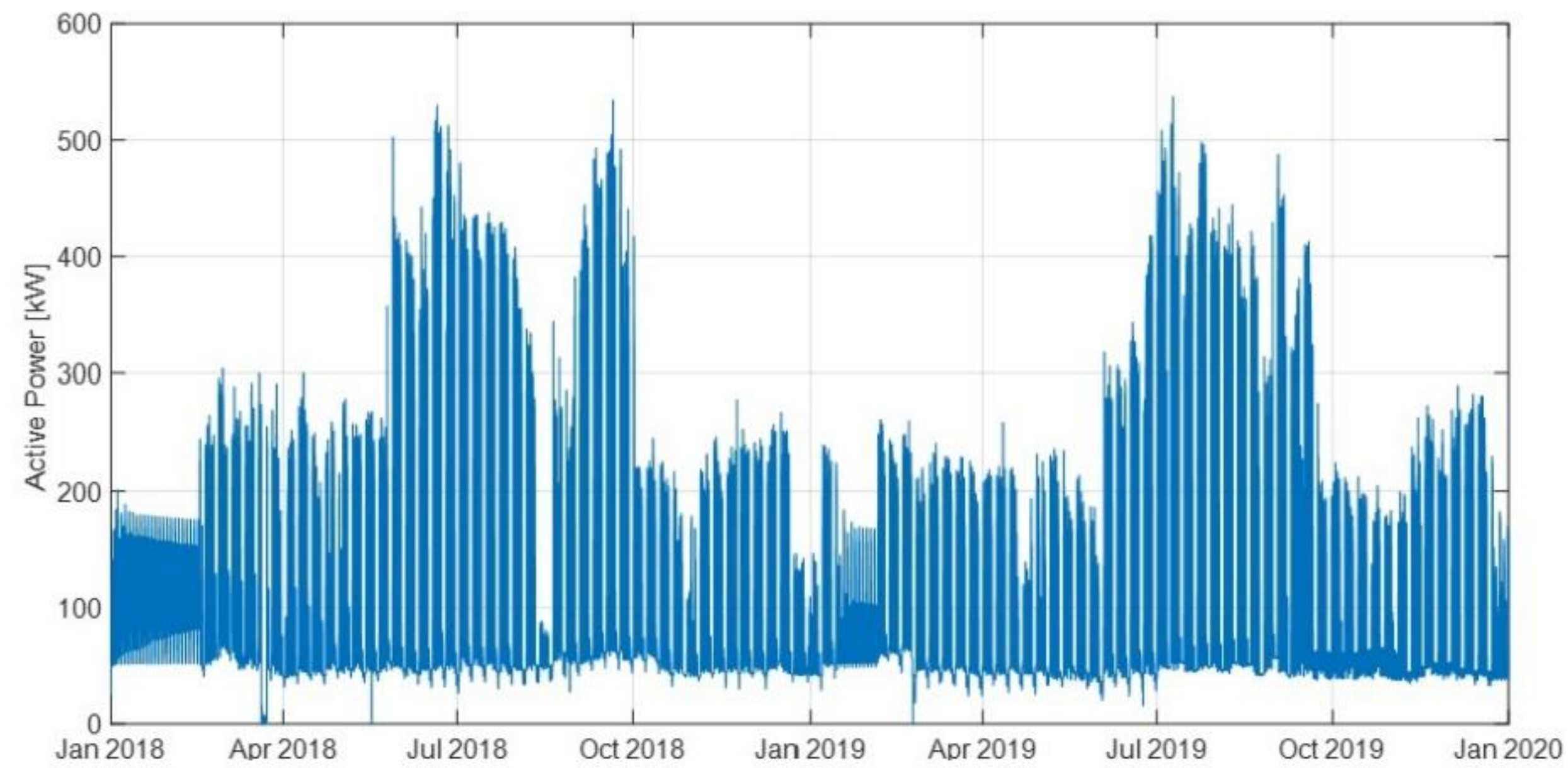
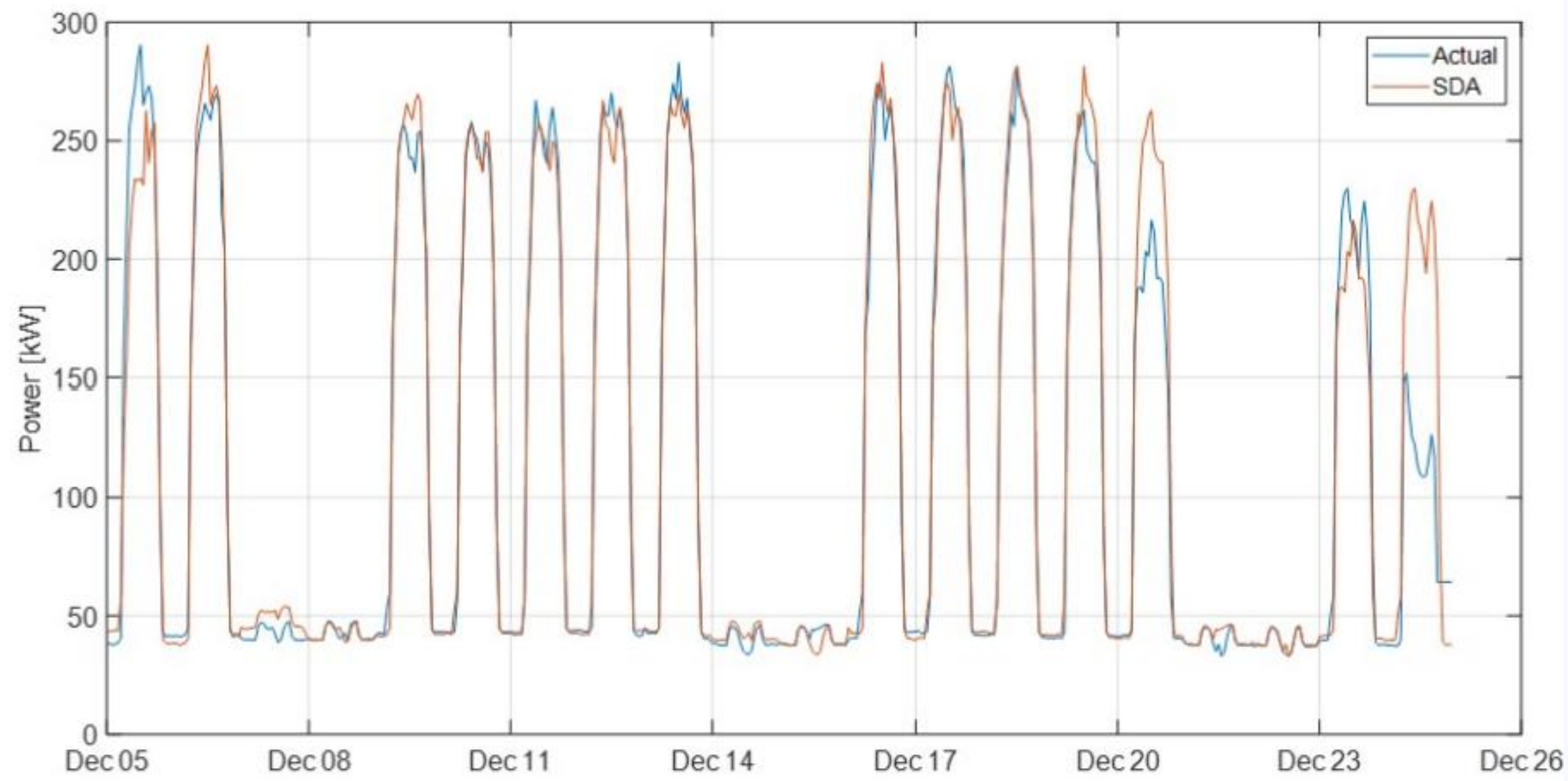
For many techniques that rely on explanatory variables, an **important step is determining which explanatory variables to use and their functional forms.**

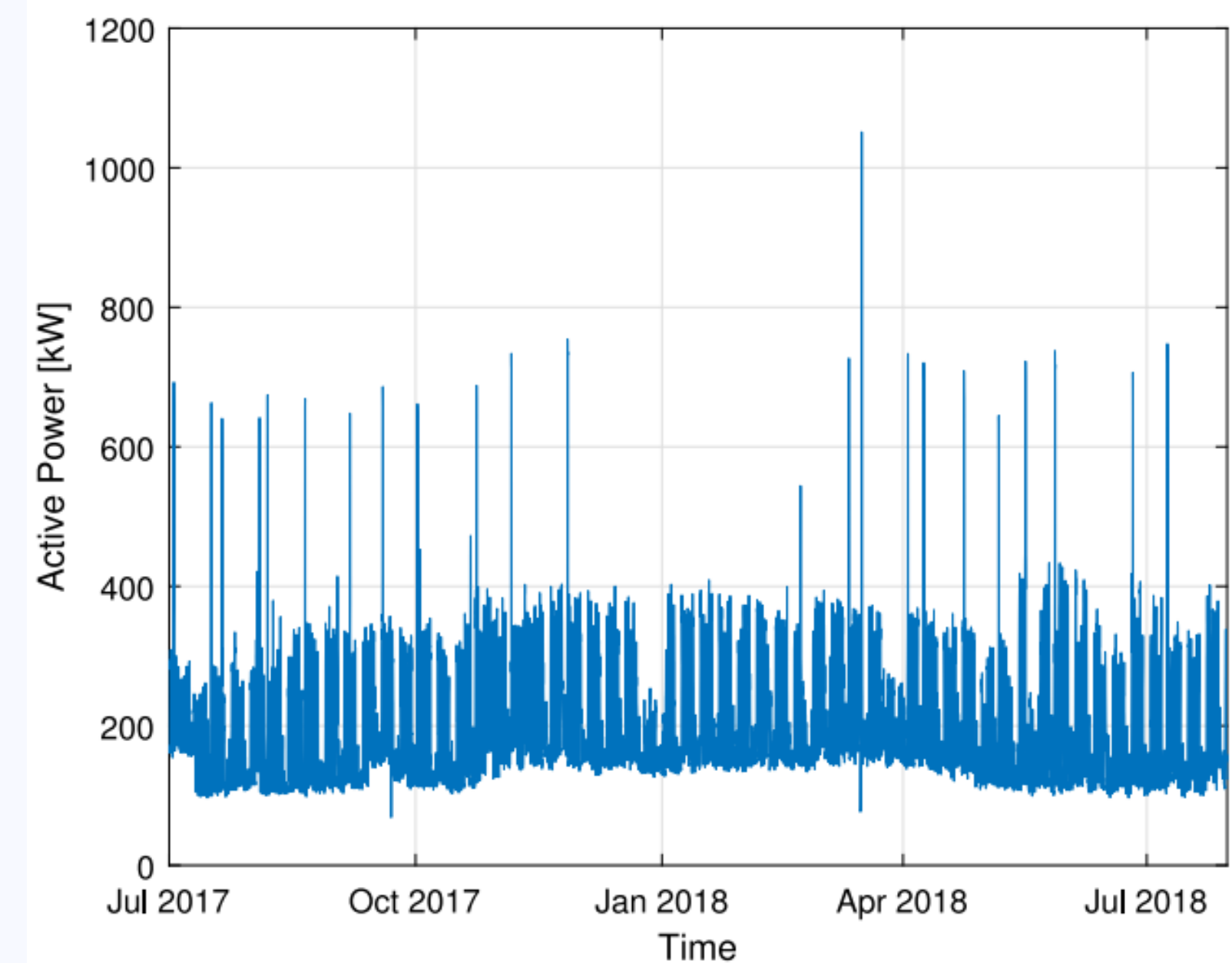
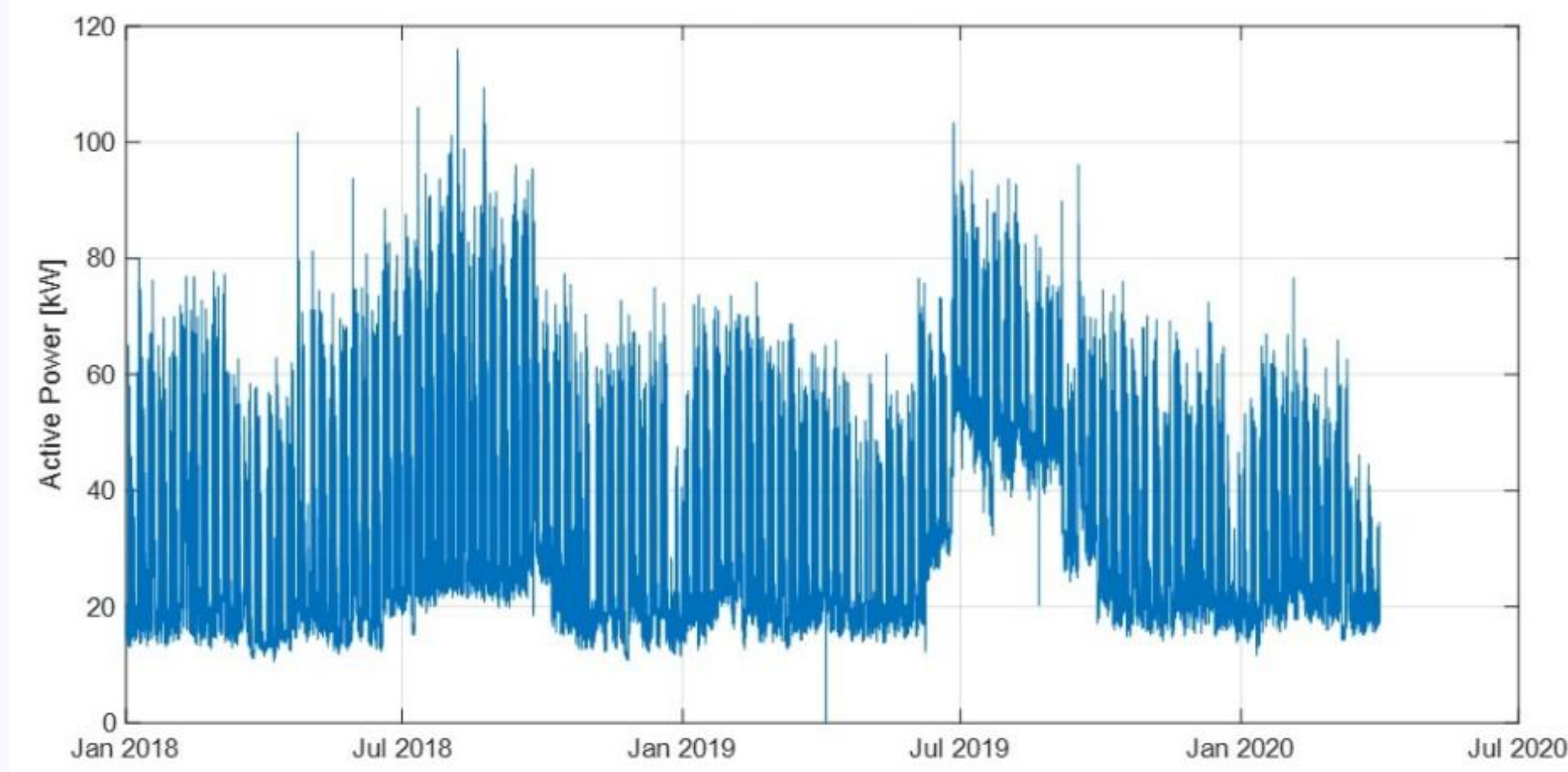
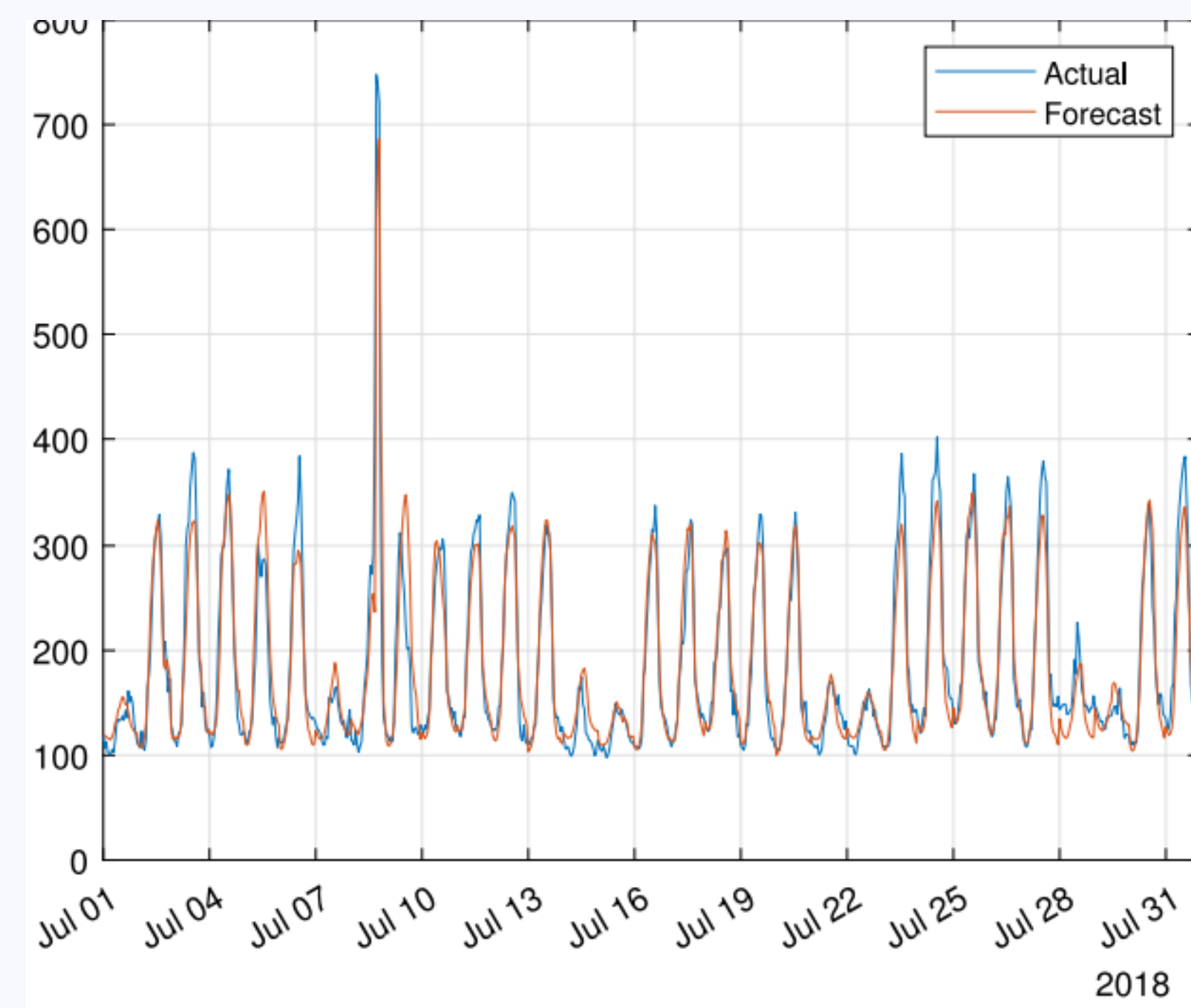
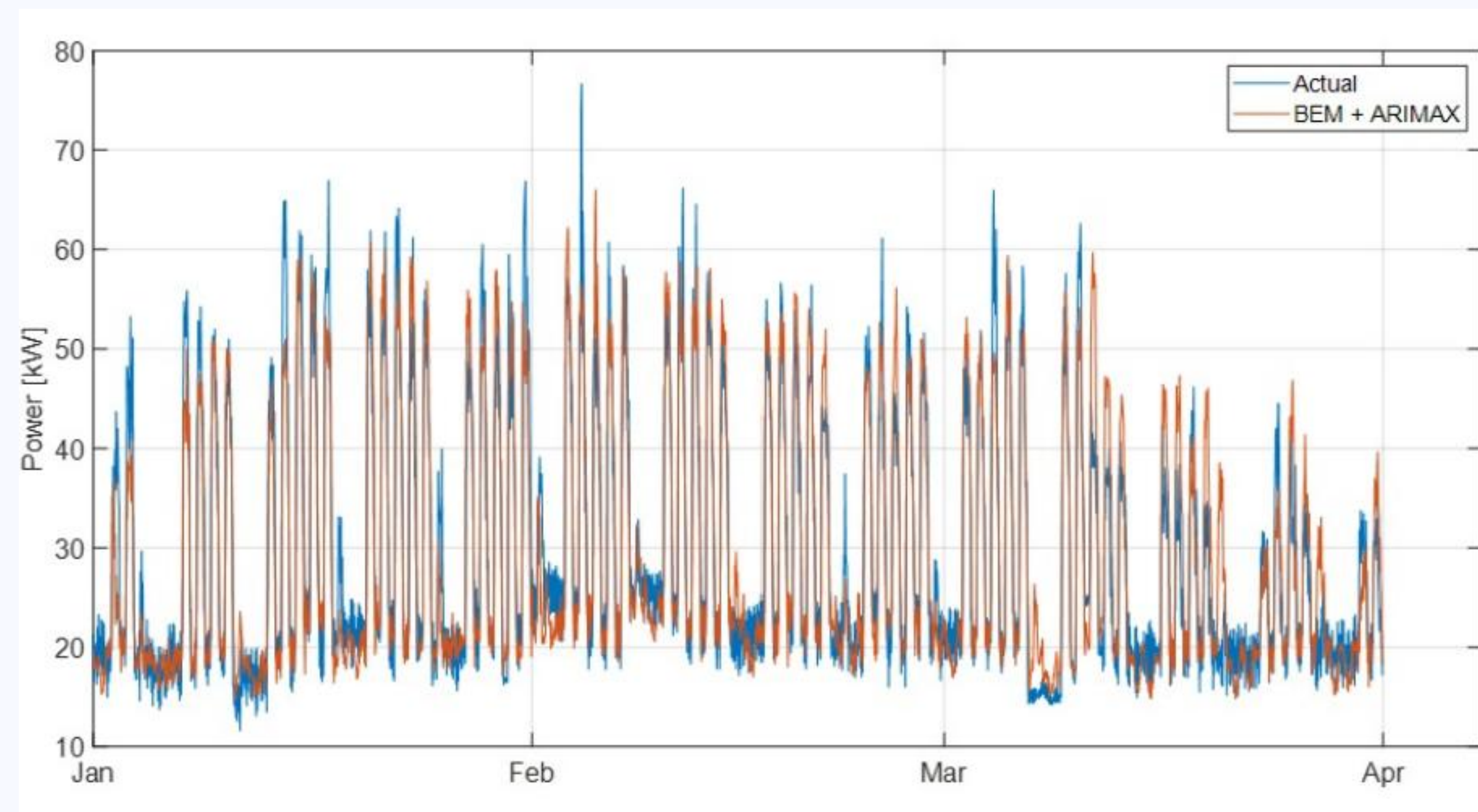
It is important to analyze the correlation among the load and all the variables in order to provide the best set of inputs to the considered technique.

Exogenous Input	Studies Number	Papers Number	Distribution Percentage
None	36	12	30.8%
Socio-economic	42	15	38.5%
Environmental	41	16	41.0%
Building	44	19	48.7%
Time Index	33	11	28.2

Ref:

C. Kuster, Y. Rezgui, M. Mourshed, “Electrical load forecasting models: A critical systematic review”, Sustainable Cities and Society, vol. 35, 2017, pp. 257-270





Short-Term Load Forecasting

State of the Art – Techniques and Methodologies

State of the art- techniques and methodologies

In this presentation the word “**technique**” refers to a group of models that fall in the same family, such as Multiple Linear Regression (MLR) models, Artificial Neural Networks (ANNs), while “**methodology**” represents the general solution framework that can be implemented with multiple techniques.

Techniques

Load forecasting techniques are typically classified into two groups: **statistical techniques** and **Artificial Intelligence (AI)** techniques.

We will briefly review four statistical techniques, namely:

- **MLR models;**
- **Semi-parametric Additive models;**
- **AutoRegressive and Moving Average (ARMA) models;**
- **Exponential Smoothing models;**

three AI techniques, namely:

- **ANN;**
- **Fuzzy Regression models;**
- **Support Vector Machines (SVMs);**

and the category composed by the **Hybrid Methods**.

Ref:

- M. Shahidehpour and T. Hong, “Load Forecasting Case Study”, Eastern Interconnection States’ Planning Council (EISPC) and National Association of Regulatory Utility Commissioners (NARUC) – Technical Report, 2015.
- T. Hong and S. Fan, “Probabilistic electric load forecasting: A tutorial review,” *International Journal of Forecasting*, vol. 32, no. 3, pp. 914–938, 2016.
- T. Hong, “Energy Forecasting: Past, Present, and Future”, *International Journal of Forecasting*, vol. 30, no. 2, pp. 357-363, 2015.
- B. Yildiz, J.I. Bilbao, and A.B. Sproul, “A Review and Analysis of Regression and Machine Learning Models on Commercial Building Electricity Load Forecasting”, *Renewable and Sustainable Energy Reviews*, vol. 73, pp. 1104-1122, 2017
- Y. Wang, Q. Chen, T. Hong, and C. Kang, “Review of Smart Meter Data Analytics: Applications, Methodologies, and Challenges”, *IEEE Transactions on Smart Grid*, vol. 10, no. 3, pp. 3125-3148, 2019.

Multiple Linear Regression (MLR) models

Regression analysis is a statistical process for estimating the relationships among variables. MLR models have been used in the literature for both STLF and LTLF.

The load or some transformation of the load is usually treated as the dependent variable, while weather and calendar variables are treated as independent variables. MLR requires the user or forecaster to specify a functional form among these variables.

Ref:

J. Luo, T. Hong, and S. Fang, “Robust Regression Models for Load Forecasting”, IEEE Transactions on Smart Grid, vol. 10, no. 5, pp. 5397-5404, 2019.

Example of a MLR technique for a 24h-ahead load forecasting

A multiple linear regression model with n predictor variables $X_1, \dots, X_n$ and a response Y , can be written as

$$Y = \beta_0 + \beta_1 X_1 + \dots + \beta_n X_n + \epsilon$$

where $\beta_0, \dots, \beta_n$ are the coefficients of the model and ϵ represents the residual (i.e. the difference between the prediction and the actual value). These errors are assumed to follow a Gaussian distribution.

In the context of load forecasting let T the time horizon and t the time index, the following equation:

$$Y_t = \beta_0 + \beta_1 X_{1,t} + \dots + \beta_{n,t} X_{n,t} + \epsilon_t, \quad t = 1, \dots, T$$

represents a MLR forecasting model with a time horizon equal to T . In this case Y_t is the forecast output for time t , while $X_{i,t}$ is the observation at time t of the i -th predictor variable.

Example of a MLR technique for a 24h-ahead load forecasting

Possible inputs:

- day of week (1-7);
- hour of the day (0-23);
- holiday/weekday (binary input: 0=holiday, 1=weekday);
- day-ahead load [kW];
- week-ahead load [kW];

Hyperparameters:

- $\beta_0, \dots, \beta_n$
- Methods for the estimation of $\beta_0, \dots, \beta_n$ (Ordinary Least Squares, robust fitting algorithms - Andrews, Bisquare, Cauchy, Fair, Huber, Logistic, Talwar and Welsch)

Semi-parametric additive models

The semi-parametric additive model is in the regression framework but designed to accommodate some non-linear relationships and serially correlated errors.

In particular, such models allow nonlinear and nonparametric terms using the framework of additive models.

In load forecasting, these generalized additive models are used to estimate the relationship between load and the explanatory variables: temperatures, calendar variables, etc.

Ref:

T. Zhao, J. Wang, and Y. Zhang, “Day-Ahead Hierarchical Probabilistic Load Forecasting with Linear Quantile Regression and Empirical Copulas”, IEEE Access, vol. 7, 80969-80979, 2019

Autoregressive Moving Average (ARMA) models

ARMA models provide a parsimonious description of a stationary stochastic process in terms of two polynomials, one an autoregression and the other a moving average.

Since the hourly electricity demand series is well-known to be nonstationary, ARIMA models, which are a generalization of ARMA models, are often used for load forecasting purposes. ARMA models can also be generalized to include exogenous variables, giving ARMAX models. It is possible to include also the seasonality effect in the SARIMA or SARIMAX models.

Exponential Smoothing

Exponential smoothing assigns weights to past observations that decrease exponentially over time, whereas in the simple moving average the past observations are weighted equally.

It does not rely on explanatory variables, meaning that it has lower data requirements than other widely used techniques such as MLR and ANN. On the other hand, these techniques can suffer cases wherein the demand is strongly driven by external variables (e.g. weather, temperature, etc.).

Input: past available load [kW].

Technique: weighted averages of past observations, with the weights decaying exponentially as the observations get older. Specifically:

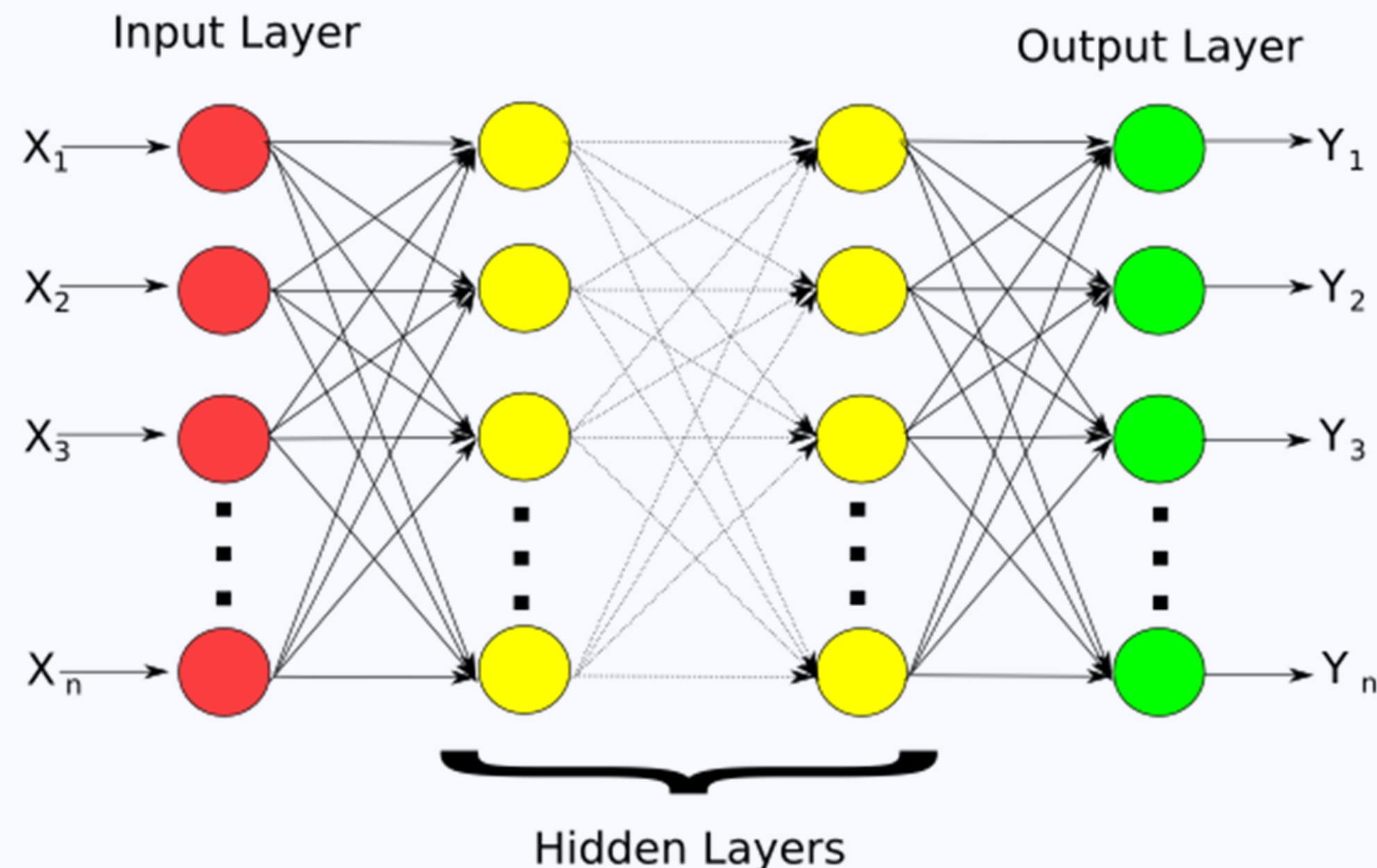
$$\hat{Y}_t = \alpha Y_{t-\gamma} + (1 - \alpha) \hat{Y}_{t-\gamma}$$

Ref:
Garulli, S. Paoletti, and A. Vicino, “Models and Techniques for Electric Load Forecasting in the Presence of Demand Response”, *IEEE Transactions on Control System Technology*, vol. 23, no. 3, 2015, pp. 1087-1097, 2017

Candidate Hyperparameters: $\alpha \in \{0.1, \dots, 0.9\}$, $\gamma \in \{24 \cdot 7, 24, \dots\}$;

Artificial Neural Networks (ANNs)

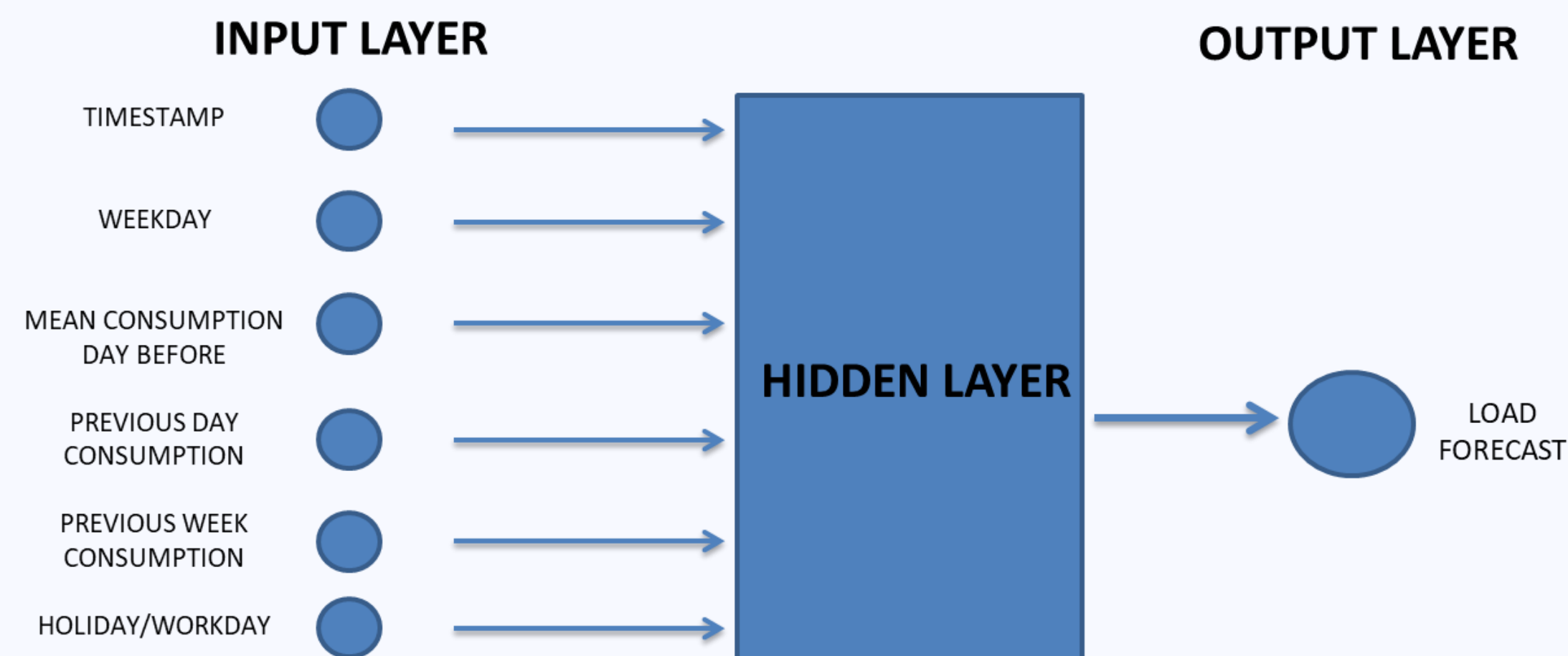
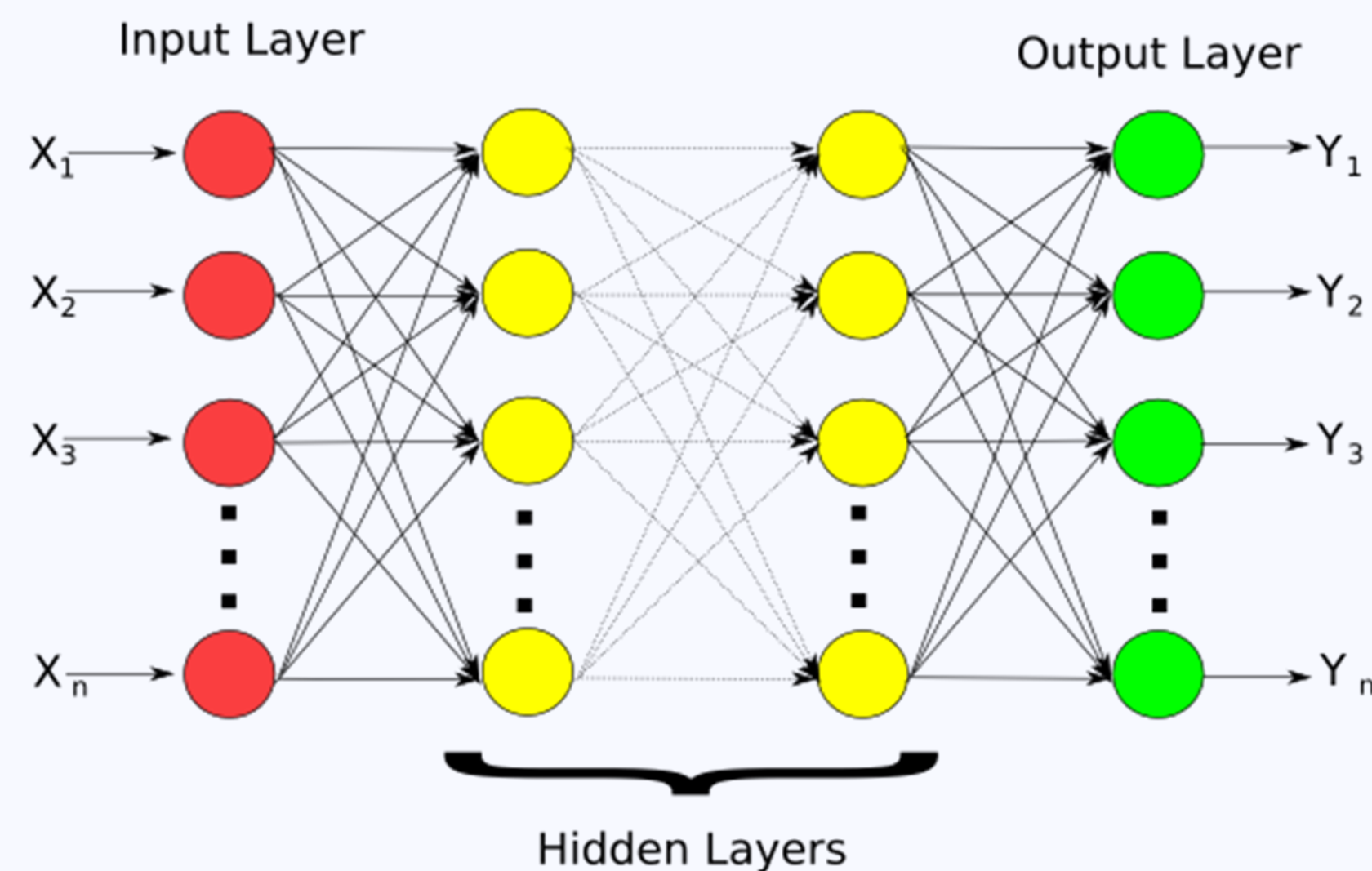
The ANN is a soft computing technique that does not require the forecaster to model the underlying physical system explicitly. In other words, the forecaster does not have to specify the functional form among the input and output variables (black box), as must be done when building MLR models. By simply learning the patterns from the historical data, a mapping between the input variables and the electricity demand can be constructed, then adopted for the prediction.



Ref:

- Bagnasco, F. Fresi, M. Saviozzi, F. Silvestro, and A. Vinci, "Electrical consumption forecasting in hospital facilities", Book chapter in *Machine Learning and Data Science in the Power Generation Industry: Best Practices, Tools, and Case Studies*, pp. 195-211, 2021.
- M. Saviozzi, S. Massucco, and F. Silvestro, "Implementation of advanced functionalities for Distribution Management Systems: Load forecasting and modeling through Artificial Neural Networks ensembles", *Electric Power Systems Research*, vol. 167, pp. 230-239, 2019

Example of an ANN technique for a 24h-ahead load forecasting



Hyperparameters:

- Architecture (Multi-Layer Perceptron)
- Transfer functions
- Number of hidden layers
- Neurons within hidden layer
- Training algorithm

15 min. int. of the day	Day of week	24 h average load [kWh]	Day ahead load [kWh]	7 days ahead load[kWh]	Weekday Holiday
1	3	77.25	73.48	61.60	1
2	3	77.27	80.52	65.92	1
3	3	77.31	82.41	66.64	1
⋮	⋮	⋮	⋮	⋮	⋮
96	3	85.18	67.32	63.48	1

Basic Ensemble Method (BEM)

There are several types of ensemble methods. In this work the **Basic Ensemble Method (BEM)** has been analyzed and implemented. The BEM output is defined as:

$$f_{BEM}(t) = \frac{1}{n} \sum_{i=1}^n f_i(t)$$

where n is the total number of ANNs and f_i are the single network outputs defined as a function of time index t .

Ref:

P. Almaleck, S. Massucco, G. Mosaico, M. Saviozzi, P. Serra, “Electrical Consumption Forecasting in Sports Venues: A proposed approach based on neural networks and ARIMAX models”, Sustainable City and Society, vol. 100, 105019

Fuzzy regression models

Fuzzy regression is introduced in order to overcome some of the limitations of linear regression, such as the vague relationship between the dependent variable and the independent variables, insufficient numbers of observations, and hard-to-verify error distributions.

The fundamental difference between the assumptions of the two techniques is related to the deviations between the observed and estimated values: linear regression assumes that these values are supposed to be errors in measurement or observations, while fuzzy regression assumes that they are due to the indefiniteness of the system structure.

Ref:

J. Luo, T. Hong, and S. Feng, “Benchmarking Robustness of Load Forecasting Models Under Data Integrity Attacks”, International Journal of Forecasting, vol. 34, pp. 89-104, 2018

Support Vector Machine (SVM)

SVMs are supervised learning models with associated learning algorithms that analyze data and recognize patterns, often being used for classification and regression analysis.

SVM has been shown to be very resistant to the problem of over-fitting, and eventually achieves good performances for solving time series forecasting problems

Ref:

X. Zhang, J. Wang, and K. Zhang, "Short-term Electric Load Forecasting Based on Singular Spectrum Analysis and Support Vector Machine Optimized by Cuckoo Search Algorithm", Electric Power System Research, vol. 146, pp. 270-285, 2017

Hybrid models

Any kind of mixture of the above-mentioned techniques is called hybrid technique.

In recent years this type of approaches have gained a lot of attention. The main goal of these strategies is to exploit the strong points of all the different techniques.

Ref:

- G. Sideratos, A. Ikonomopoulos, N. Hatziargyriou, “A Novel Fuzzy-based Ensemble Model for Load Forecasting using Hybrid Deep Neural Networks”, *Electric Power Systems Research*, vol. 178, 2020
- J. Zhang, Y. Wei, D. Li, Z. Tan, J. Zhou, “Short Term Electricity Load Forecasting Using a Hybrid Model”, *Energy*, vol. 158, 2018, pp. 774-781

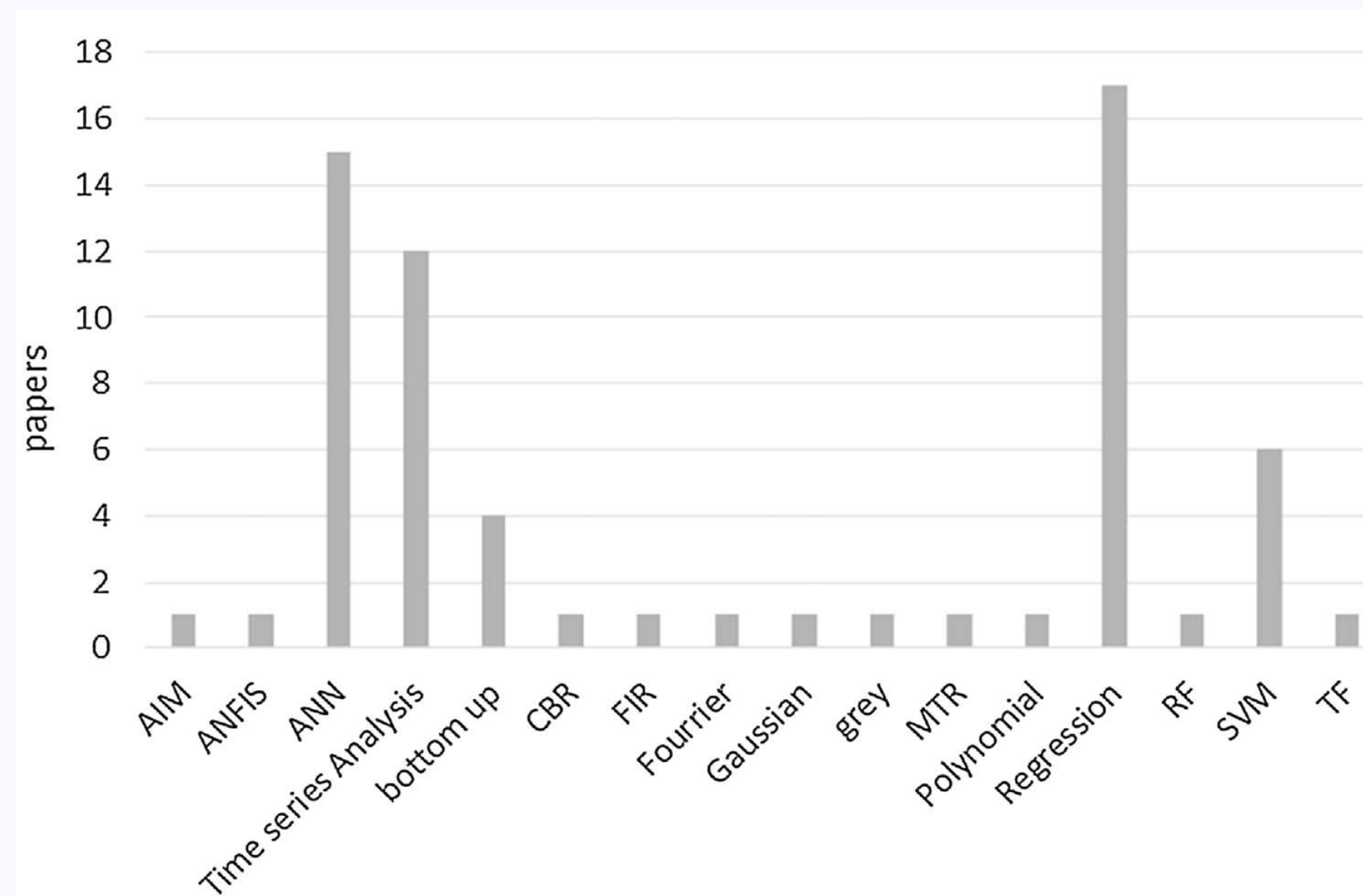
Trivial techniques

Very useful as simple benchmark or in green field situations (no available data).

Examples:

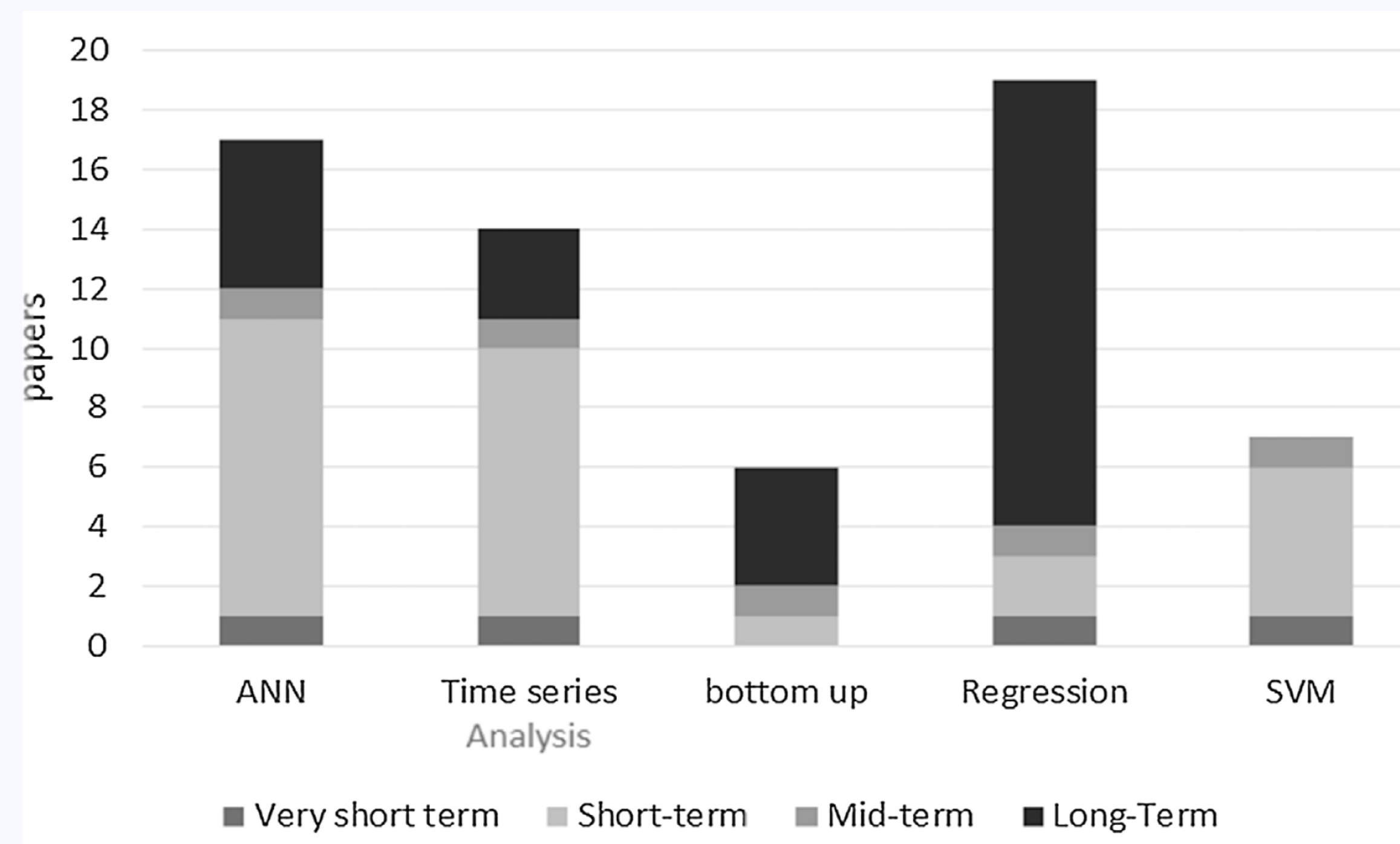
- Day ahead load is considered as forecast;
- Week ahead load is considered as forecast;
- Similar Day Approach (SDA): the idea of this approach is to find a day in the historical data that is similar to the day being forecasted.

Techniques distribution



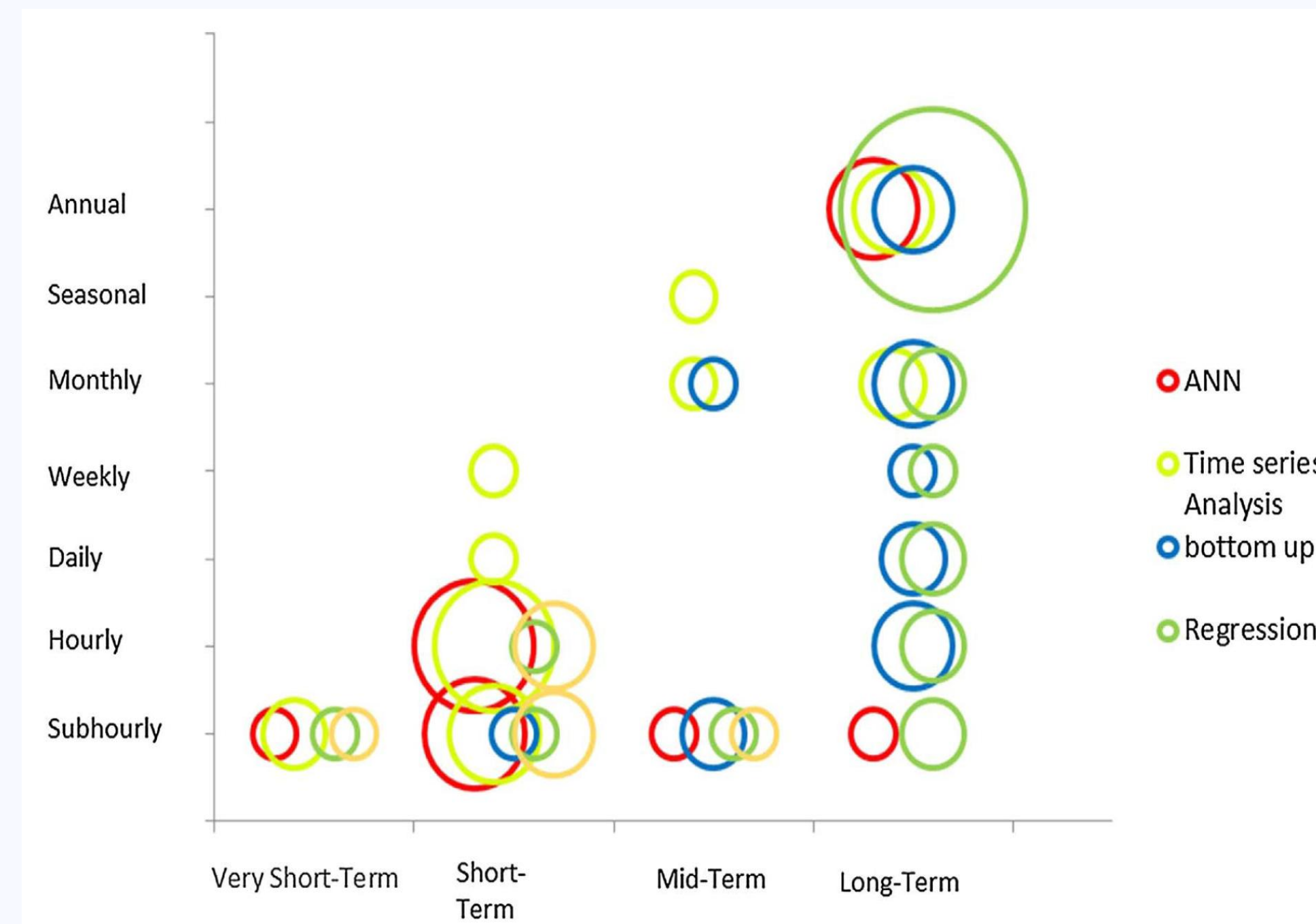
Ref: C. Kuster, Y. Rezgui, M. Mourshed, "Electrical load forecasting models: A critical systematic review", Sustainable Cities and Society, vol. 35, 2017, pp. 257-270

Technique vs Time horizon



Ref: C. Kuster, Y. Rezgui, M. Mourshed, "Electrical load forecasting models: A critical systematic review", Sustainable Cities and Society, vol. 35, 2017, pp. 257-270

Technique vs Time horizon vs Resolution



Ref: C. Kuster, Y. Rezgui, M. Mourshed, "Electrical load forecasting models: A critical systematic review", Sustainable Cities and Society, vol. 35, 2017, pp. 257-270

The myth of the best technique

It is very important for researchers and practitioners that a universally best technique does not exist. It is the data and jurisdictions that determine what technique we should use, rather than the other way around.

We should always understand the business needs first, then analyze the data, and usually go through a trial-and-error process, to figure out which is the best technique for a specific dataset in a specific jurisdiction.

Note that the forecasting error may also differ significantly for different utilities, different zones within a utility, and different time periods.

Ref:

- T. Hong and S. Fan, "Probabilistic electric load forecasting: A tutorial review," *International Journal of Forecasting*, vol. 32, no. 3, pp. 914–938, 2016
- S. Massucco, G. Mosaico, M. Saviozzi, P. Almaleck, M. Santarelli and P. Serra, "Development and Selection of Load Forecasting Techniques for Non-Residential Buildings," 2024 18th International Conference on Probabilistic Methods Applied to Power Systems (PMAPS), Auckland, New Zealand, 2024

Methodologies

- **Similar Day Method:** The idea of the similar day method is to find a day in the historical data that is similar to the day being forecasted. The similarity is usually based on day of the week, season of the year, and weather patterns.
- **Feature Extraction**
- **Feature Selection**
- **Training algorithms**
- **Hierarchical Forecasting**

Short Term Load Forecasting

Key Performance Indicators (KPIs)

Key Performance Indicators (KPIs)

Several criteria are utilized as evaluation indices for load forecasting techniques to ascertain the accuracy of predicting actual load values.

Various researchers have relied on different statistical metrics to quantify the accuracy of their models, and over time, new statistical metrics are emerging, including those for probabilistic load forecasting.

- **Mean Absolute Error (MAE)**

$$MAE = \sum_{t=1}^N |Y_t - \hat{Y}_t|$$

Ref:

T. Hong and S. Fan, "Probabilistic electric load forecasting: A tutorial review," International Journal of Forecasting, vol. 32, no. 3, pp. 914–938, 2016

Key Performance Indicators (KPIs)

- **Mean Absolute Percentage Error (MAPE)**

$$MAPE = \frac{1}{N} \sum_{t=1}^N \frac{|Y_t - \hat{Y}_t|}{Y_t} \cdot 100\%$$

- **Root Mean Square Error (RMSE)**

$$RMSE = \sqrt{\frac{1}{N} \sum_{t=1}^N (Y_t - \hat{Y}_t)^2}$$

Key Performance Indicators (KPIs)

- **Root Relative Squared Error (RRSE)**

$$RRSE = \frac{\sqrt{\sum_{t=1}^N (Y_t - \hat{Y}_t)^2}}{\sqrt{\sum_{t=1}^N (Y_t - \bar{Y}_t)^2}}$$

- **Skill Score (SS)**

$$SS = 1 - \frac{|KPI_{proposed}|}{|KPI_{reference}|}$$

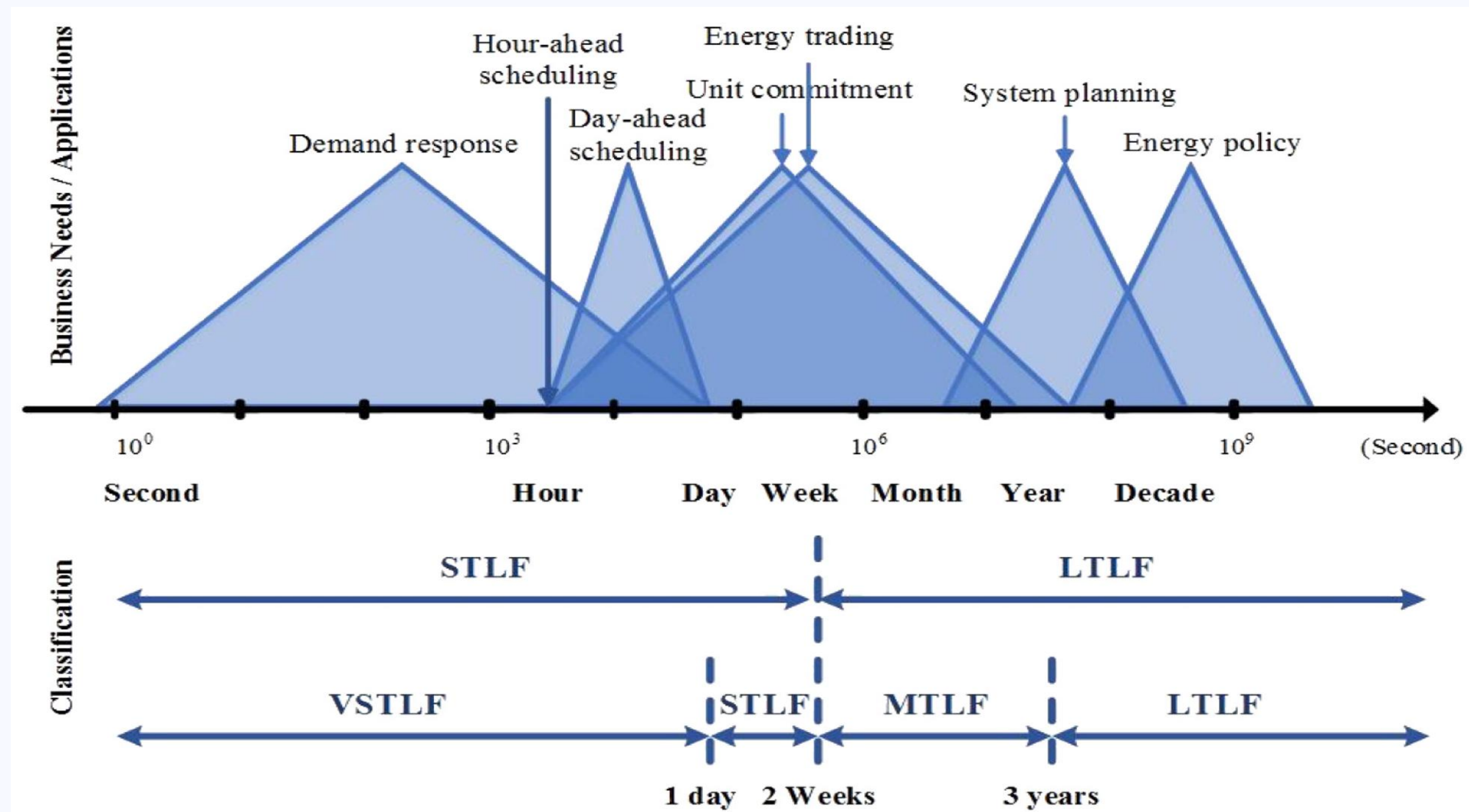
- **Coefficient of Variation (CV)**

$$CV = \frac{RMSE}{\bar{Y}_t}$$

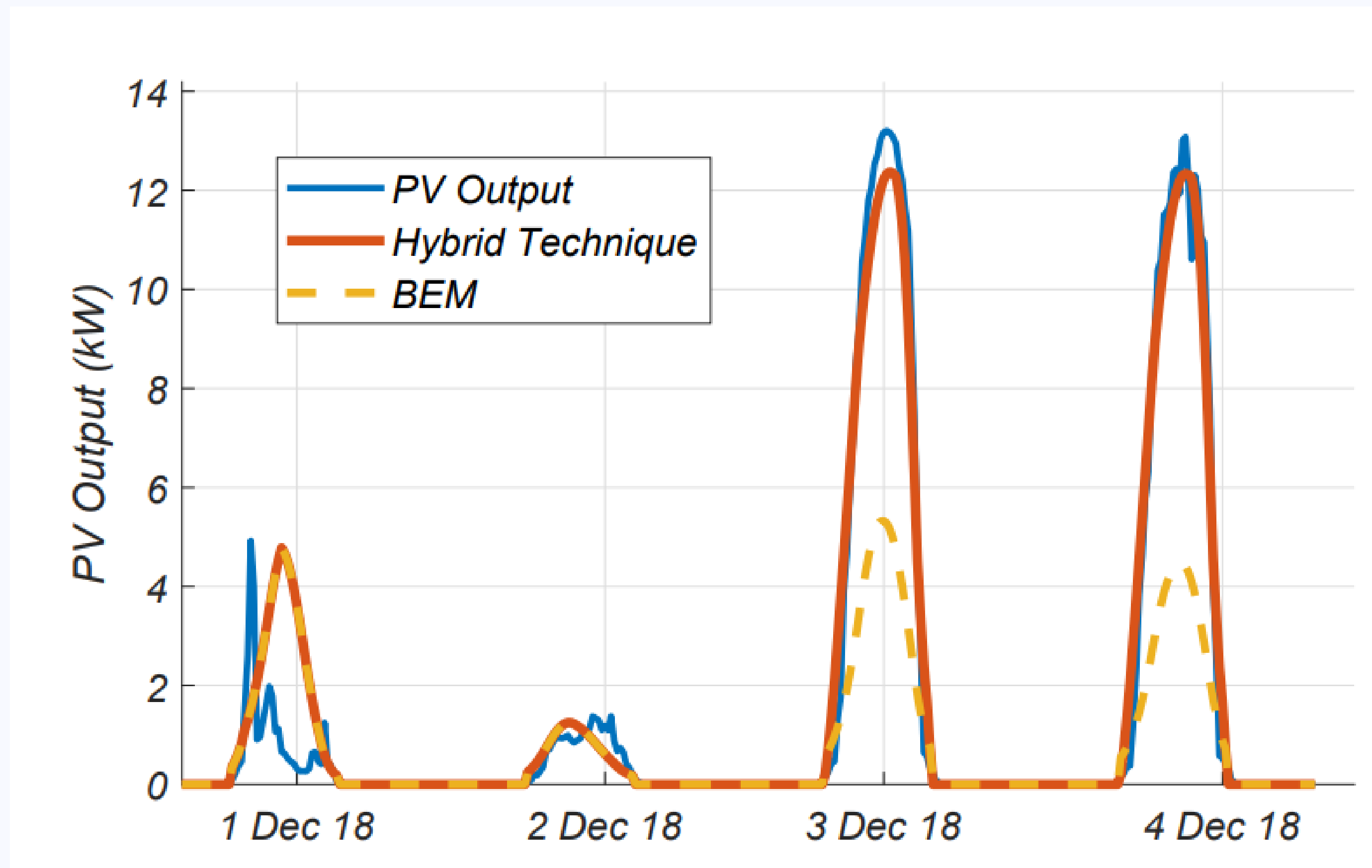
Very Short/Short-Term PV Forecasting

Basic Concepts

Classification of PV forecasting problem

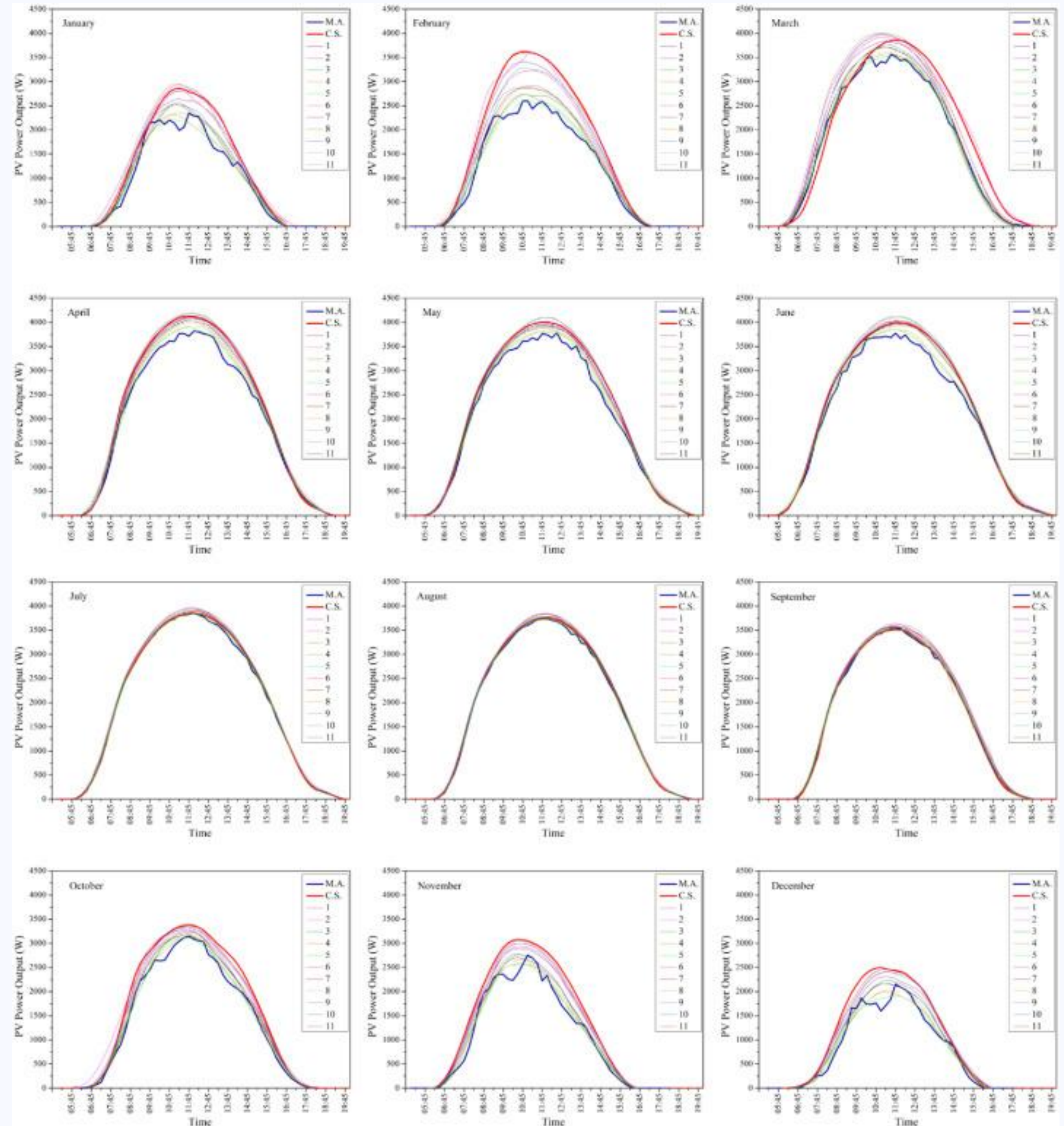


Type of days





Type of days



Fundamental parameters/inputs

Parameter	Value
Peak Power (P_{peak})	20 kW _p
Efficiency of the Inverter (η_{inv})	0.971
Temperature Coefficient (β_c)	$-0.045 \frac{1}{^{\circ}\text{C}}$
Tilt Angle (β)	30°
Latitude	44.4141°
Longitude	8.9221°
Local Time Zone	15°
Orientation regarding S, positive W	30°
Number of Inverters	2
Inverters Nominal power	12.5 kW
Number of Panels	108
Nominal Power of each Panel	180 W
Surface of each Panel	1.3 m ²
Coefficient of Degradation (ω_{DEG})	1

Key inputs

- Time of day
- Day of the year
- Historical production data
- Plant technical data
- Weather variables (e.g., solar irradiance, temperature, humidity, wind speed, etc.)

Very Short/Short-Term PV Forecasting

State of the Art

State of the art - Classification

A first distinction is made between **direct and indirect forecasting methods**.

In an indirect forecasting approach, the solar irradiance is forecasted and then exploited in commercial PV simulation software to predict the PV power generation.

Direct methodologies on the other hand aim to directly predict the PV output.

Ref:

R. Ahmed, V. Sreeram, Y. Mishra, M.D. Arif, "A review and evaluation of the state-of-the-art in PV solar power forecasting: techniques and optimization", Renew Sustain Energy Rev, 124, 2020.

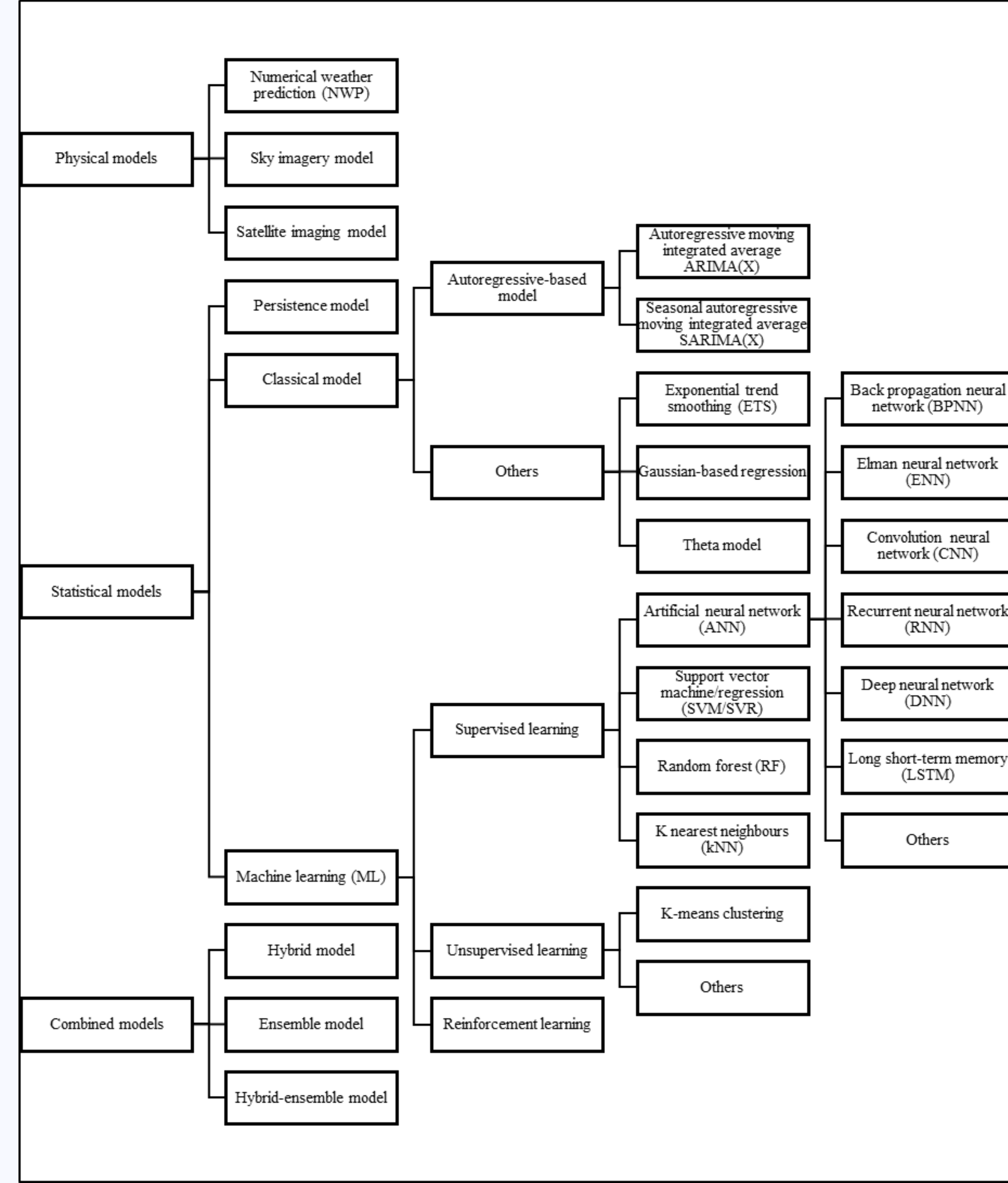
State of the art - Classification

A more common classification, within direct methods, divides the PV forecasting techniques into 3 categories:

- **Physical models;**
- **Statistical models;**
- **Combined or Hybrid models.**

Ref:

T. N. Nguyen and F. Müsgens, “What drives the accuracy of PV output forecasts?”, *Applied Energy*, Volume 323, 119603, 2022



Physical models

Definition:

Physical models (also known as PV performance, parametric, or “white box” models) use mathematical and physical principles to forecast photovoltaic (PV) power output.

Key Inputs:

- **Irradiance** (affected by sun position, temperature, humidity, albedo, shading, clouds)
- **Module Temperature** (influenced by irradiance, ambient temperature, wind speed)

Advantages:

- Do **not require historical data**
- Can generate inputs for **hybrid statistical models**

Main Types:

- **Numerical Weather Prediction (NWP)**
- **Sky Imagery**
- **Satellite Imaging**

Ref:

- S. Pelland, J. Remund, J. Kleissl, T. Oozeki, and K. De Brabandere, “Photovoltaic and solar forecasting: State of the art,” tech. rep., IEA, 2013. Report
- C. Monteiro, T. Santos, L.A. Fernandez-Jimenez, I.J. Ramirez-Rosado, and M.S. Terreros-Olarte, “Short-term power forecasting model for photovoltaic plants based on historical similarity”, *Energies*, 6, pp. 2624-2643, 2013

Numerical Weather Prediction (NWP)

Purpose:

NWP tools provide atmospheric condition forecasts over specific time scales, enhancing renewable energy prediction models.

Model Types:

- **Global Models:** Simulate atmospheric behavior on a global scale
- **Mesoscale Models:** Focus on localized regions (e.g., countries, regions)

Applications:

- Combined with AI models for **short-term PV power prediction**
- Used to build **weather parameter databases**
- Enable **scenario-based forecasting** using historical data and pattern mining

Ref:

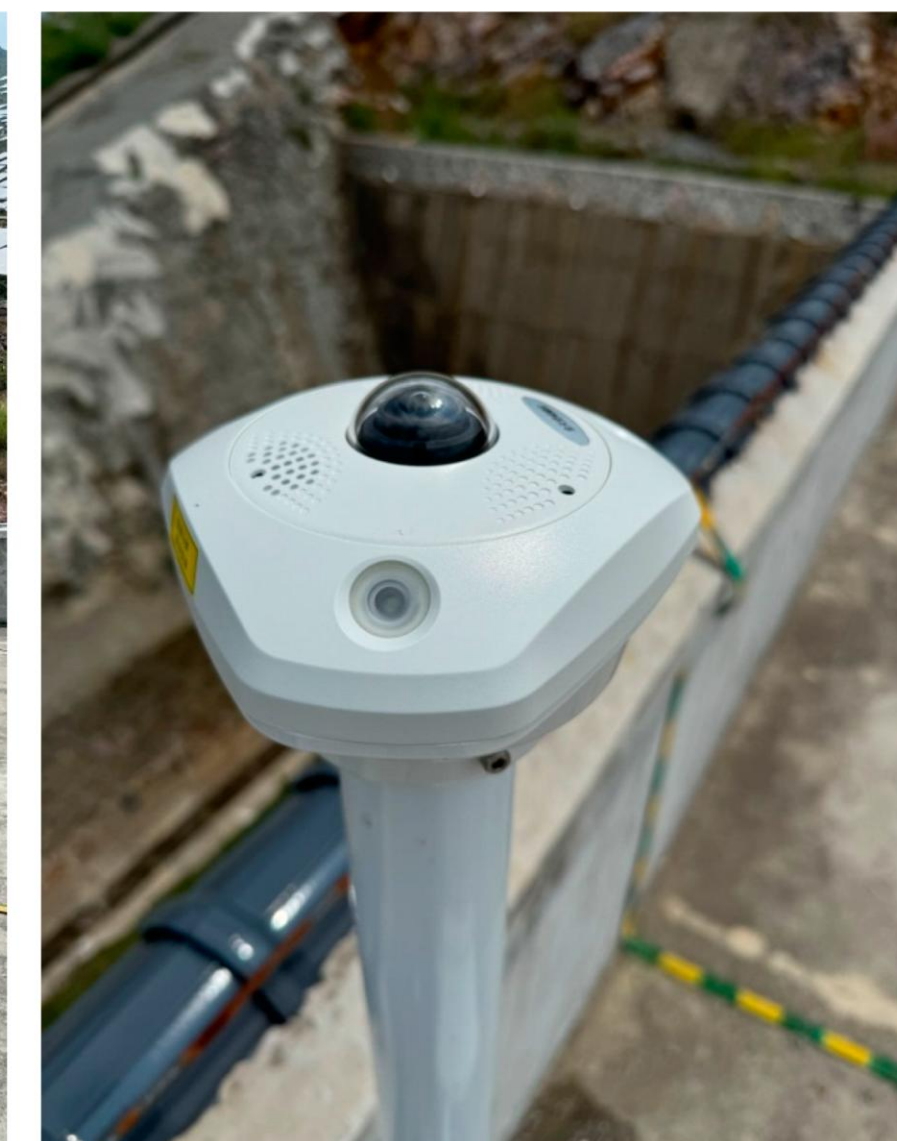
C. Monteiro, T. Santos, L.A. Fernandez-Jimenez, I.J. Ramirez-Rosado, and M.S. Terreros-Olarte, “Short-term power forecasting model for photovoltaic plants based on historical similarity”, *Energies*, 6, pp. 2624-2643, 2013

Sky Imagery

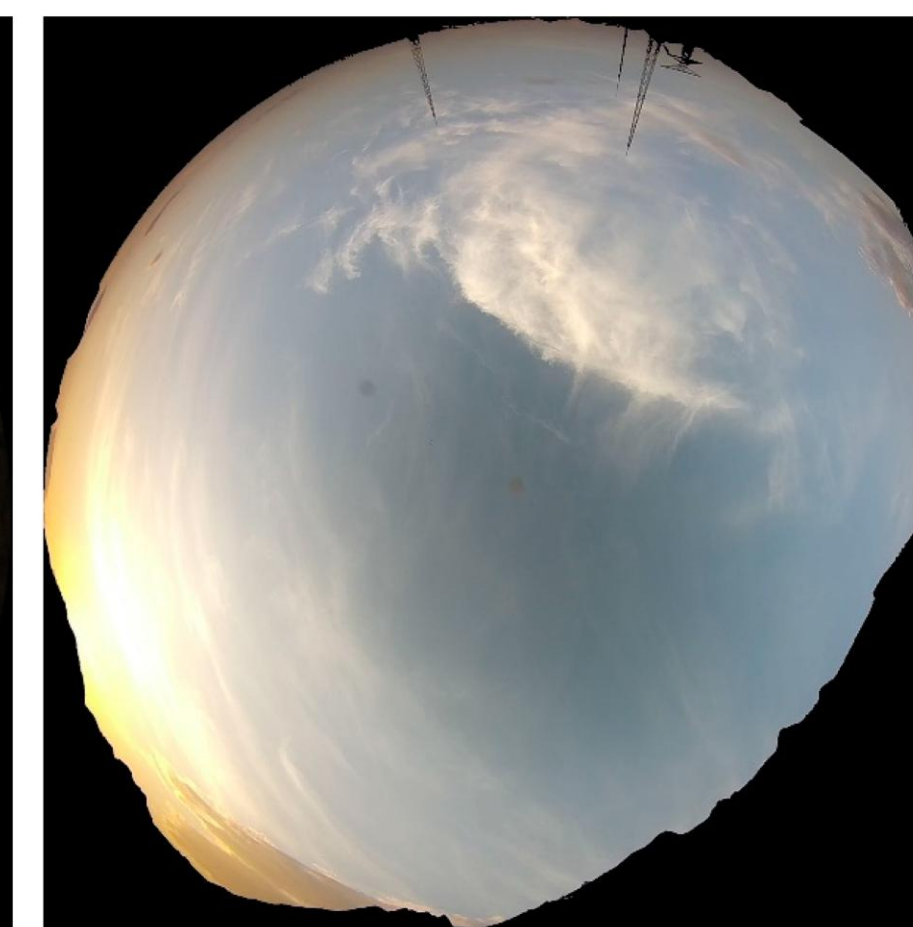
A **sky imager is a digital camera** that captures high-quality images of the sky from horizon to horizon. It is commonly used for cloud detection, measuring cloud height above the ground, and determining cloud motion.

Most existing sky imagers are designed to capture a hemispheric view of the sky. Their built-in cameras are either oriented upwards to obtain a direct image or downwards to capture reflections from a spherical mirror.

Ref:
S. Sobri, S. Koohi-Kamali, N. A. Rahim, "Solar photovoltaic generation forecasting methods: A review", *Energy Conversion and Management*, vol. 156, pp. 459-497, 2018



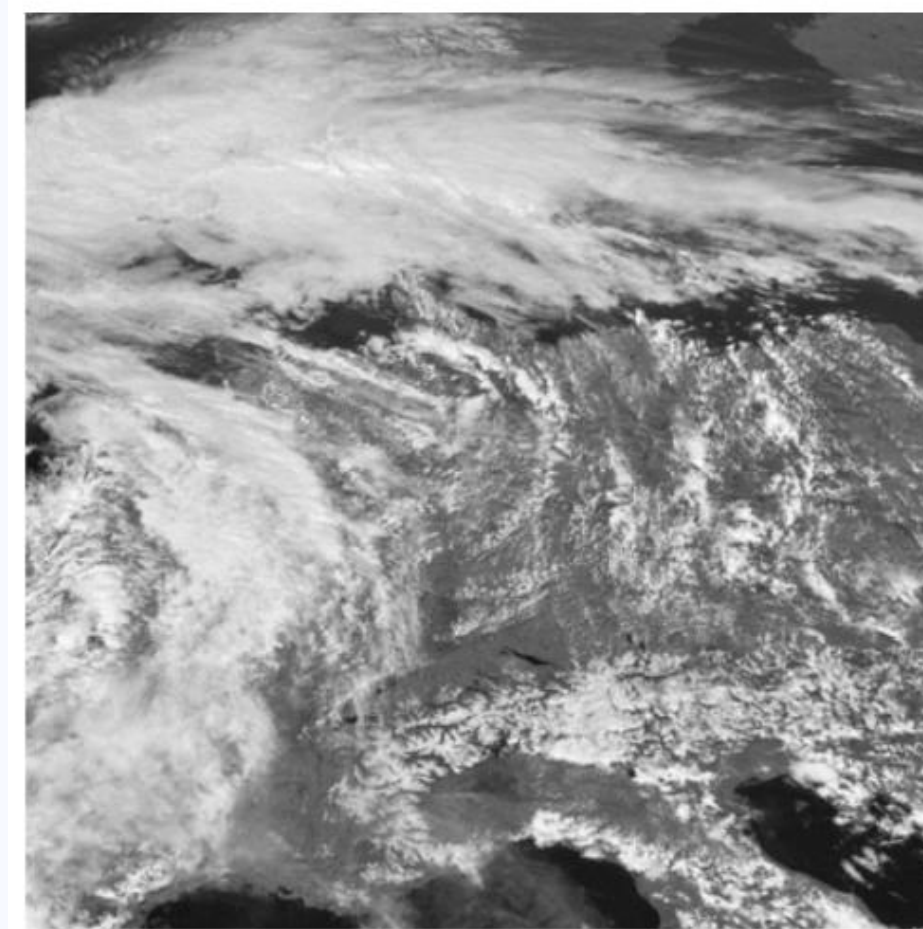
(a) Segmentation of sky boundaries



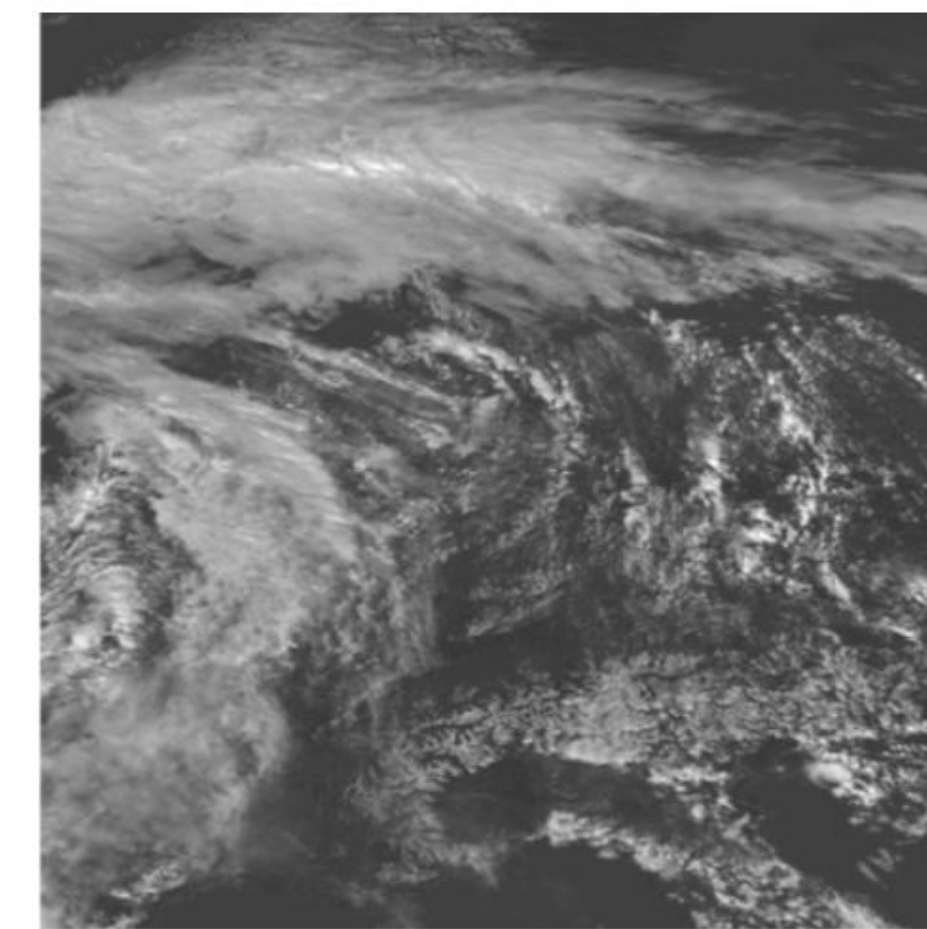
(b) Extraction of sky region

Satellite imaging

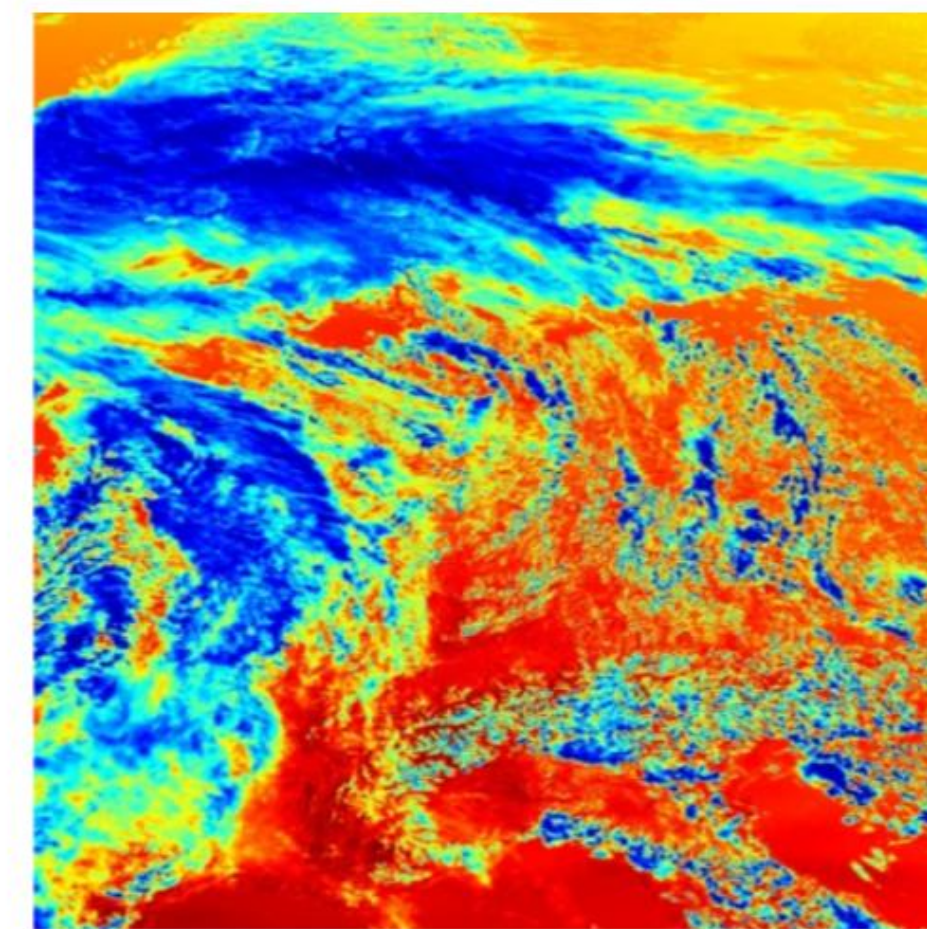
The concepts of **satellite imaging** closely resemble those of sky imagery models. Cloud patterns are discerned from visible and infrared images captured by satellite-based sensors orbiting overhead.



Meteosat-10 raw image



Cloud index: comparison between actual and clear sky for each pixel



GHI: global horizontal irradiation at ground level

Ref:
S. Sobri, S. Koohi-Kamali, N. A. Rahim, "Solar photovoltaic generation forecasting methods: A review", *Energy Conversion and Management*, vol. 156, pp. 459-497, 2018

Statistical models

Overview:

- Widely used for **short-term forecasting**
- Based on **historical weather and PV production data**
- Do **not require internal system details**
- **Data-driven:** learns patterns from past data

Advantages:

- Can outperform physical models
- Corrects **systematic input errors**
- Ideal for plants with **rich historical datasets**

Model Categories:

- **Persistence Models**
- **Classical Models** (ARMA, ARIMA, SARIMA, SARIMAX, Gaussian regression, Exponential Smoothing)
- **Machine Learning Techniques** (Supervised learning, Unsupervised learning, Reinforcement learning)

Ref:

J. Antonanzas, N. Osorio, R. Escobar, R. Urraca, F.J. Martinez-de-Pison, and F. Antonanzas-Torres, “Review of photovoltaic power forecasting”, *Solar Energy*, vol. 136, pp. 78-111, 2016

Persistence model

Definition:

- Also known as the **naive** or **elementary** model

Assumes that PV output at time t on day $d+1$ equals output at the same time t on day d

Key Characteristics:

- **Simplest statistical model**
- Requires only **historical PV output data**
- No weather or system inputs needed

Use Case:

- Commonly used as a **benchmark**
- New models are typically evaluated against it
- Only models that **outperform persistence** are considered for further development

Ref:

J. Antonanzas, N. Osorio, R. Escobar, R. Urraca, F.J. Martinez-de-Pison, and F. Antonanzas-Torres, "Review of photovoltaic power forecasting", Solar Energy, vol. 136, pp. 78-111, 2016

Classical models

Core Techniques:

- **AR Models** (AutoRegressive)
- **SARIMA** (Seasonal ARIMA)
- **SARIMAX** (SARIMA with exogenous variables)

Advantages of Extended Models:

- Better handling of **non-stationary data**
- Improved **forecasting accuracy** over basic AR models

Other Classical Approaches:

- **Gaussian Regression**
- **Exponential Trend Smoothing (ETS)**
- **Theta Model**

Key Point:

These models are well-established, interpretable, and effective for structured time series forecasting.

Ref:

- P. Bacher, H. Madsen, and H. Nielsen, "Online short-term solar power forecasting", *Sol. Energy*, 83, pp. 1772-1783, 2009
- Y. Li, Y. Su, and L. Shu, "An ARMAX model for forecasting the power output of a grid connected photovoltaic system", *Renew. Energy*, 66, pp. 78-89, 2014
- Y. Chu, B. Urguhart, S. Gohari, H. Pedro, J. Kleissl, and C. Coimbra, "Short-term reforecasting of power output from a 48 MWe solar PV plant", *Sol. Energy*, 112, pp. 68-77, 2015
- H.T.C. Pedro and C.F.M. Coimbra, "Assessment of forecasting techniques for solar power production with no exogenous inputs", *Sol. Energy*, 86, pp. 2017-2028, 2012
- M. Bouzerdoun, A. Mellit, and A. Massi Pavan, "A hybrid model (SARIMA-SVM) for short-term power forecasting of a small-scale grid-connected photovoltaic plant", *Sol. Energy*, 98, pp. 226-235, 2013

Machine Learning (ML)

Strengths:

- Handles **complex, non-linear relationships**
- **Self-adaptive** and capable of strong **inference**
- Suitable for **multi-input/multi-output** scenarios

Challenges:

- More **complex** than classical methods
- Requires **higher computational resources**

ML Categories:

- **Supervised Learning** – trained on labeled data
(Examples: Artificial Neural Networks (ANN), K-Nearest Neigh Neighbors, Support Vector Machines/Regression (SVM/SVR), Random Forest)
- **Unsupervised Learning** – uses unlabeled data
- **Reinforcement Learning** – agent learns by interacting with the environment to maximize rewards

Ref:

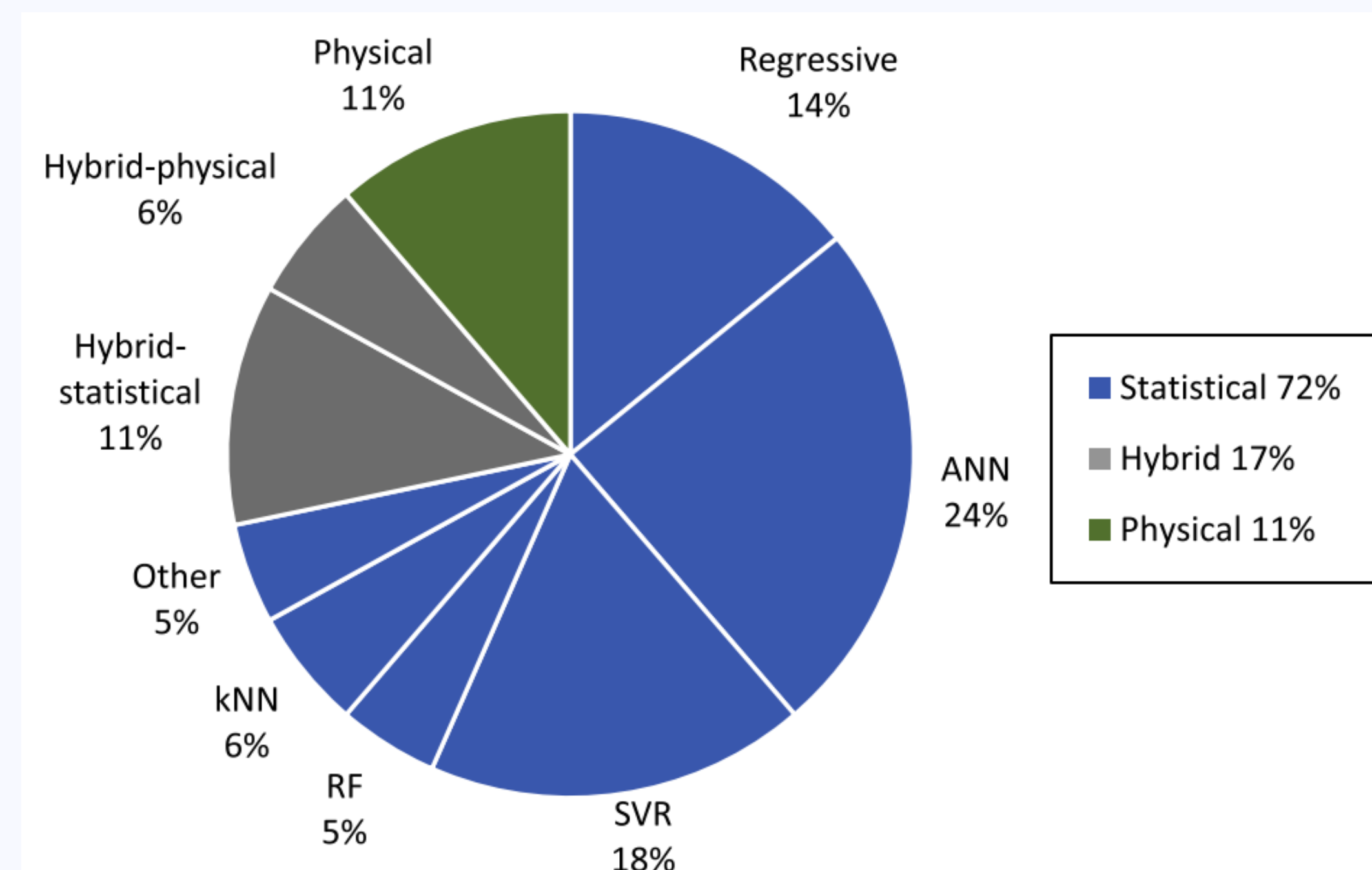
R. Rajagukguk, R. Ramadhan, and H. Lee, “A review on deep learning models for forecasting time series data of solar irradiance and photovoltaic power”, *Energies*, 13(24), 6623, 2020

Artificial Neural Networks (ANNs)

The ANNs are the most used machine learning techniques in solar power forecasting.

A reason behind this popularity is the ANN capability to capture the high nonlinearity of PV production.

In this context, also variables characterized by a poor linear correlation with PV output can help the ANN to improve the forecasting accuracy.



Ref:

- J. Antonanzas, N. Osorio, R. Escobar, R. Urraca, F.J. Martinez-de-Pison, and F. Antonanzas-Torres, "Review of photovoltaic power forecasting", *Solar Energy*, vol. 136, pp. 78-111, 2016
- M. G. De Giorgi, P. M. Congedo, and M. Malvoni, "Photovoltaic power forecasting using statistical methods: Impact of weather data," *IET Science, Measurement and Technology*, vol. 8, pp. 90-97, 2014

Hybrid Models

Definition: Combined models integrate multiple forecasting techniques and include:

Hybrid Models: Combine physical + statistical methods ("grey box")

Ensemble Models: Aggregate outputs from multiple models

Hybrid-Ensemble Models: Merge both hybrid and ensemble approaches

Key Features:

- **Higher complexity** in design and parameterization
- **Flexible architecture:** outputs of one model can feed into another
- **Improved accuracy** by correcting systematic errors or enhancing input features

Insight: Hybrid models often **outperform standalone physical or statistical models**, especially in complex forecasting scenarios.

Ref:

- T. Asrari, X. Wu, and B. Ramos, "A hybrid algorithm for short-term solar power prediction – sunshine state case study," *IEEE Transactions on Sustainable Energy*, vol. 8, pp. 582–591, 2017
- E. Ogliari, A. Dolara, G. Manzolini, and S. Leva, "Physical and hybrid methods comparison for the day ahead pv output power forecast," *Renewable Energy*, vol. 113, pp. 11–21, 2017
- S. Massucco, G. Mosaico, M. Saviozzi, and F. Silvestro, "A Hybrid Technique for Day-Ahead PV Generation Forecasting Using Clear-Sky Models or Ensemble of Artificial Neural Networks According to a Decision Tree Approach", *Energies*, 12, 1298, 2019.

Final discussion

- On average, **hybrid models demonstrate more robust performance than other models**, suggesting they will play a crucial role in improving PV output forecasting.
- **Results on machine learning (ML) models are varied.** Although they do not yet outperform other models on average, their substantial improvement over time makes them a promising candidate for future use.
- Forecast horizons exhibit a positive correlation with forecast errors, indicating greater difficulty in forecasting longer horizons.
- Forecast errors reported for a single day are considerably lower than those for longer test sets, hinting at possible “cherry-picking” of errors and underscoring the need for standardized and transparent practices in model validation. Based on our findings, we recommend a test set length of at least one year for robust model validation in future studies.
- Indirect forecasting achieves higher accuracy than direct forecasting, especially for very short-term horizons and when ML models are employed.

Ref:

T. N. Nguyen and F. Müsgens, “What drives the accuracy of PV output forecasts?”, Applied Energy, Volume 323, 119603, 2022

Short Term PV Forecasting

Key Performance Indicators (KPIs)

Key Performance Indicators (KPIs)

Below are the formulas for the error metrics extracted from the several studies on PV output forecasts [47]. These are the standard formulas utilized by all reviewed papers:

$$RMSE_{avg}(\%) = \frac{\sqrt{\frac{1}{N} \sum_{t=1}^N (Y_t - \hat{Y}_t)^2}}{\bar{Y}} \cdot 100$$

$$NRMSE_{norm}(\%) = \frac{\sqrt{\frac{1}{N} \sum_{t=1}^N (Y_t - \hat{Y}_t)^2}}{\bar{Y}} \cdot 100$$

$$NRMSE_{installed}(\%) = \frac{\sqrt{\frac{1}{N} \sum_{t=1}^N (Y_t - \hat{Y}_t)^2}}{Y_{installed/peak}} \cdot 100$$

$$NMAE_{avg}(\%) = \frac{\frac{1}{N} \sum_{t=1}^N |Y_t - \hat{Y}_t|}{\bar{Y}} \cdot 100$$

Ref:

D. Yang, J. Kleissl, C.A. Gueymard, H.T. Pedro, and C.F. Coimbra, "History and trends in solar irradiance and PV power forecasting: a preliminary assessment and review using text mining", *Sol Energy*, 168, pp. 60-101, 2018

Key Performance Indicators (KPIs)

$$NMAE_{installed}(\%) = \frac{\frac{1}{N} \sum_{t=1}^N |Y_t - \hat{Y}_t|}{Y_{installed/peak}} \cdot 100 \quad MAPE_{avg}(\%) = \frac{\frac{1}{N} \sum_{t=1}^N |Y_t - \hat{Y}_t|}{|\bar{Y}_t|} \cdot 100$$

$$MAPE_{installed}(\%) = \frac{\frac{1}{N} \sum_{t=1}^N |Y_t - \hat{Y}_t|}{Y_{installed/peak}} \cdot 100$$

where *avg*, *installed*, and *norm* indicate the methods of error normalization (using average power, installed capacity or peak power, and normalized data, respectively), N is the total number of forecast points in the forecasting period, $\hat{Y}_t$ and Y_t represent the forecast and measured values of the PV output at the time step t , $\bar{Y}$ stands for the mean value of PV output, $Y_{installed/peak}$ indicates the installed capacity of the PV plant or the peak power achieved by the plant

Ref:

D. Yang, J. Kleissl, C.A. Gueymard, H.T. Pedro, and C.F. Coimbra, "History and trends in solar irradiance and PV power forecasting: a preliminary assessment and review using text mining", *Sol Energy*, 168, pp. 60-101, 2018

Thanks for the attention

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